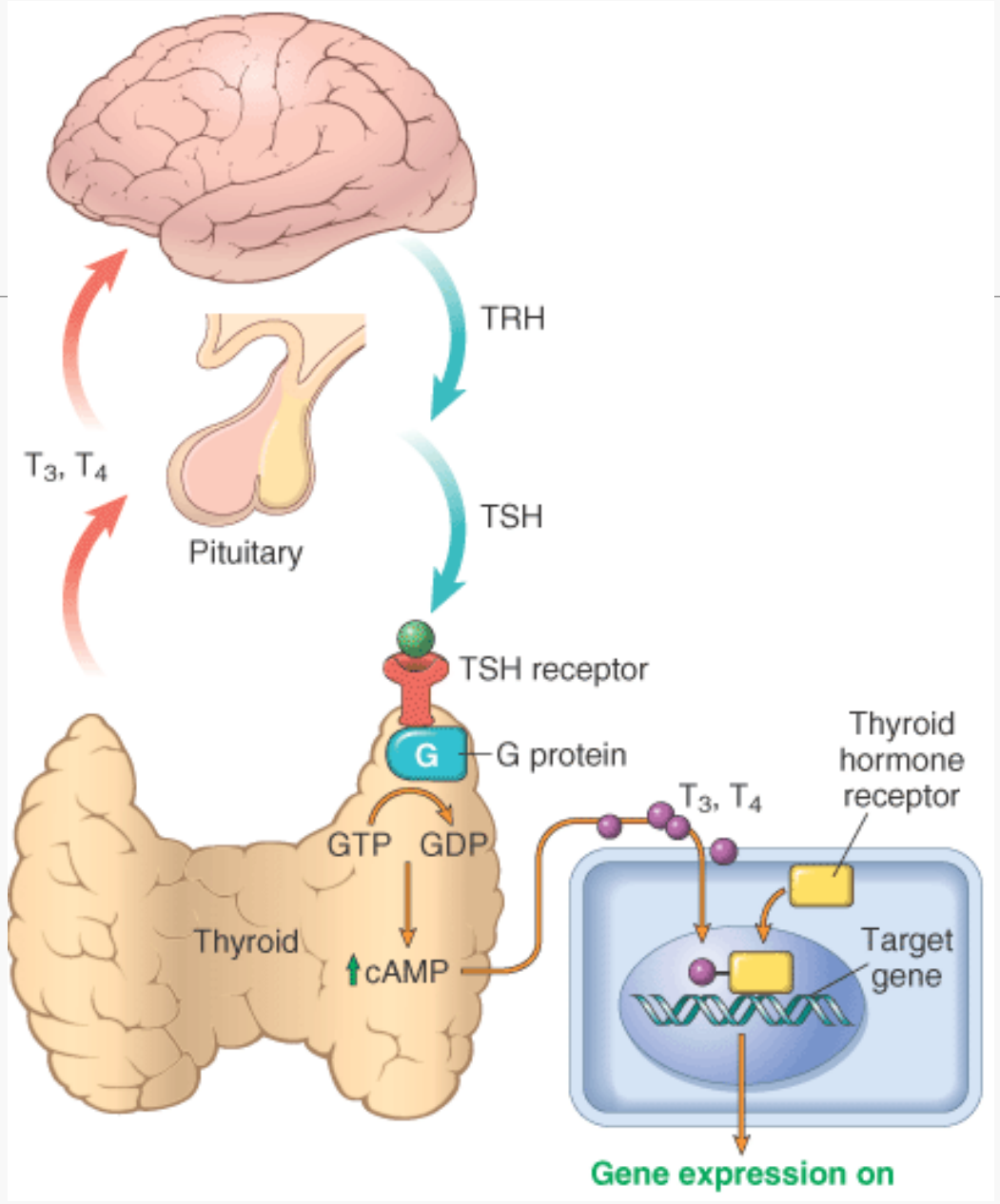
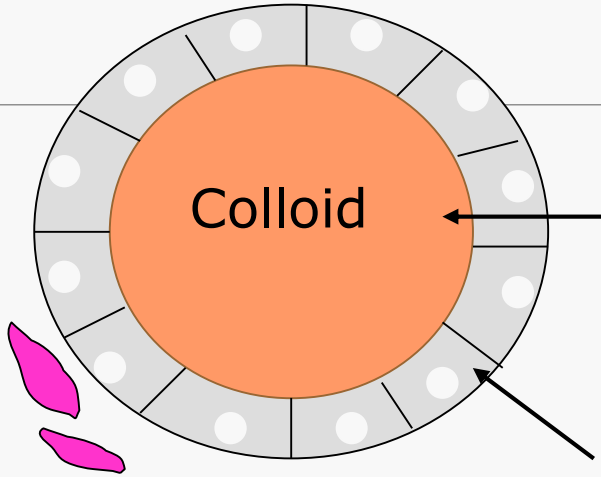
Pathology of thyroid gland and parathyroid gland

ศ.พญ.ปารมี ทองสุกใส
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tparamee@gmail.com


TRH
TSH
T_3, T_4
Pituitary
TSH receptor
G
G protein
GTP
GDP
Thyroid
cAMP
T_3, T_4
Thyroid hormone receptor
Target gene
Gene expression on

Thyroid follicles



75% Thyroglobulin

25% Iodine, thyronine,

Follicular cells

↓

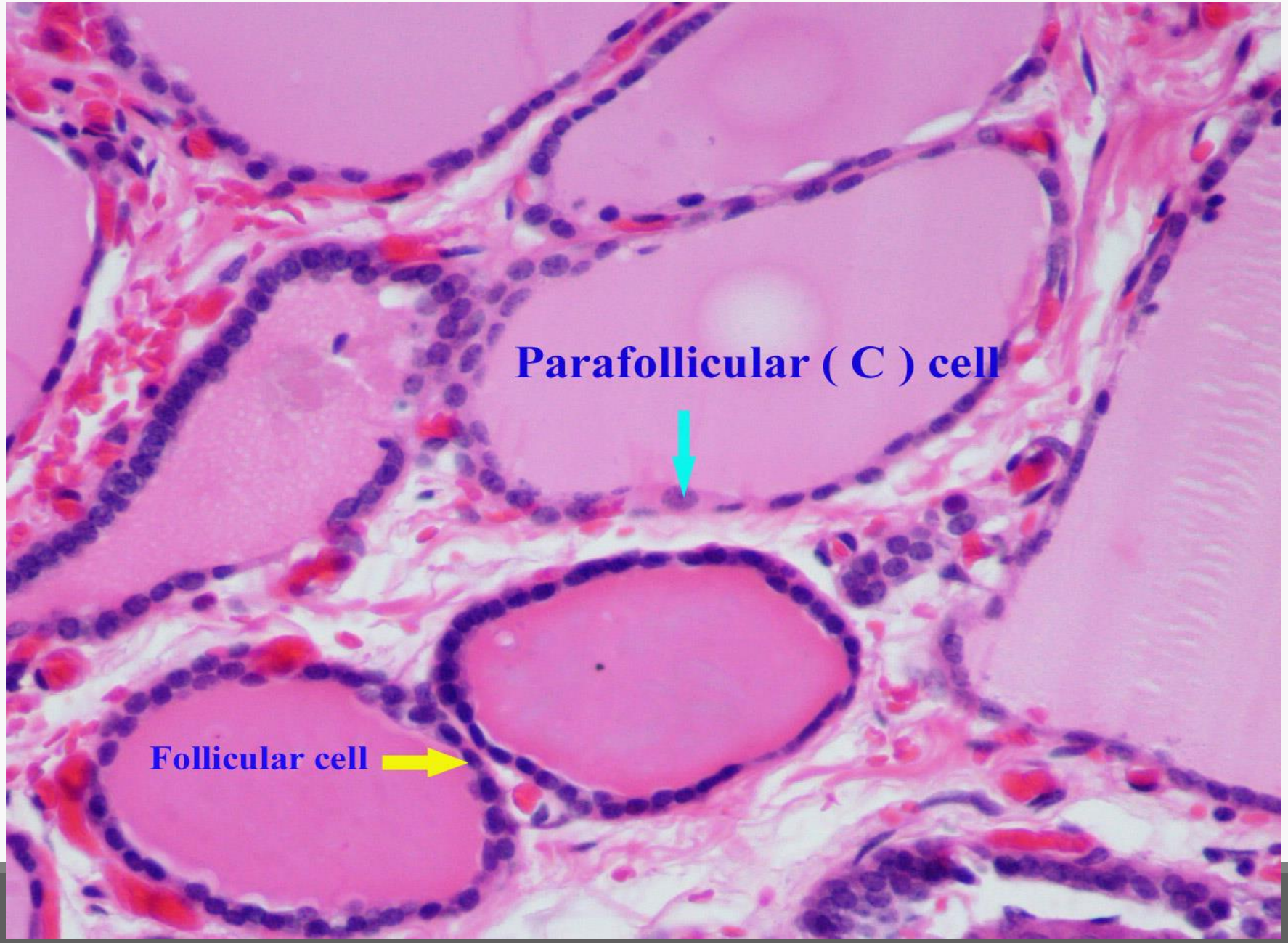
Thyroid hormone

Parafollicular cells

↓

Calcitonin

Normal thyroid



Common presentation of thyroid diseases

- Goiter (คอพอก) : น้ำหนัก > 40 กรัม (ไม่ขึ้นกับ function)
- Hyperthyroidism
- Hypothyroidism
- Euthyroid
- Nodule

Common thyroid diseases

- Graves' disease
- Hashimoto's thyroiditis
- Subacute thyroiditis
- Multinodular goiter
- Tumor
  - Benign
  - Malignant

Grave's disease

- Other names: toxic goiter, primary hyperthyroidism, primary hyperplasia, diffuse hyperplasia

---

Age: 20-40 years

Female: male 4:1

Pathogenesis: autoimmune disease

Antibody (IgG) to TSH receptor

→ hyperplasia and hormone production

Antibody to eye muscle

→ inflammation, edema, accumulation of mucopolysaccharide

Antibody to skin → accumulation of mucopolysaccharide

Grave's disease

ลักษณะอาการแสดงสำคัญ 3 อย่าง

- Hyperthyroidism
- Exopthalmos
- Dermatopathy

ผู้ป่วยอาจมีอาการแสดงทั้ง 3 อย่าง หรือพบเพียง 1-2 อย่าง และอาการใดนำมาก่อนก็ได้

Laboratory

- Increased T3 or T4,
- Decreased TSH and
- Positive thyroid receptor antibodies (thyroglobulin, thyroid peroxidase, sodium iodide symporter and TSH receptor)

Grave's disease

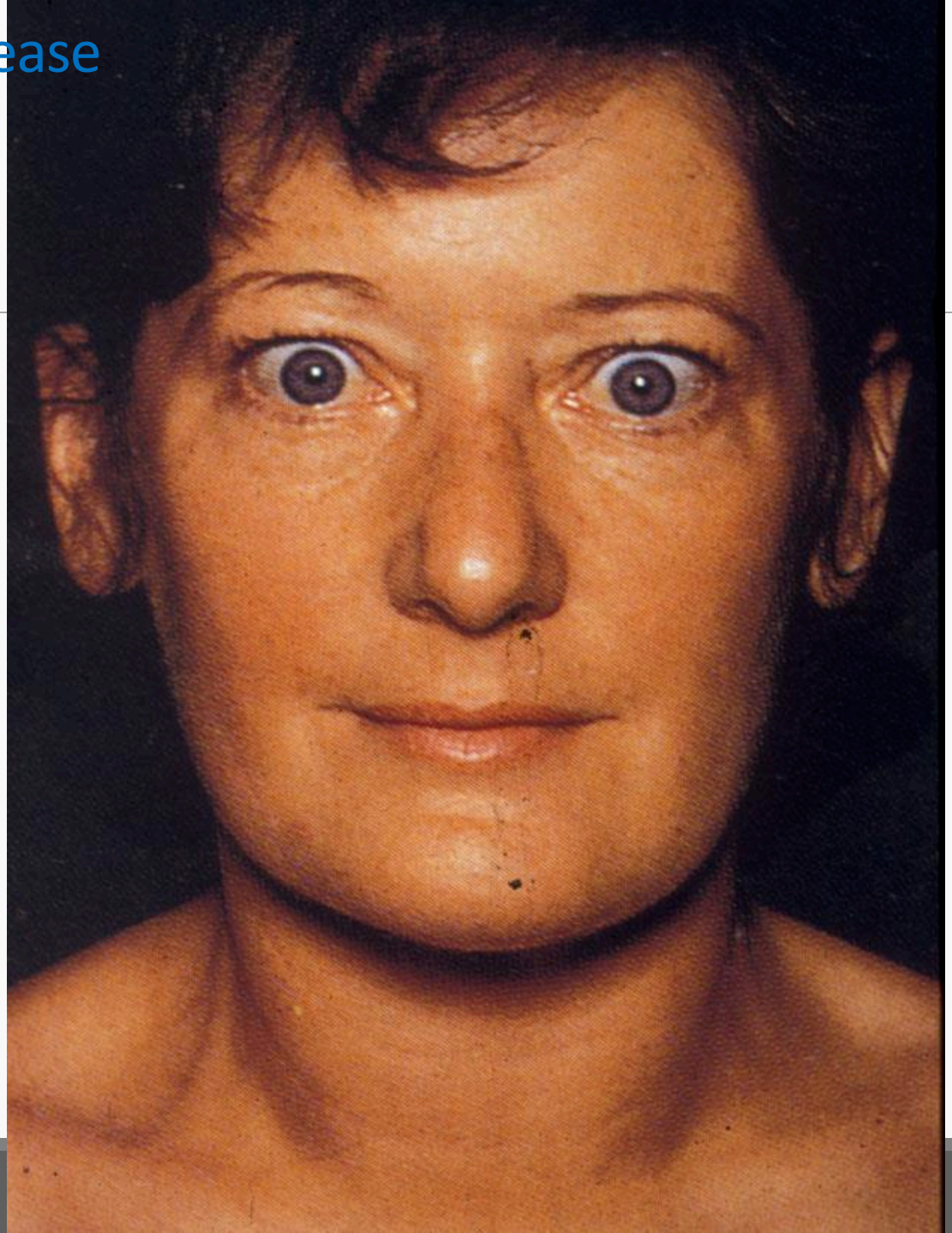
Common symptoms

- nervousness, anxiety, irritability, mood swings
- hyperactivity
- feeling tired all the time
- sensitivity to heat
- difficulty sleeping
- muscle weakness
- diarrhea
- persistent thirst

Common signs

- a swelling in the neck caused by an enlarged thyroid gland (goiter)
- fast heart rate (palpitations)
- warm skin and excessive sweating
- patchy hair loss or thinning
- weight loss – often despite an increased appetite

Graves’ disease



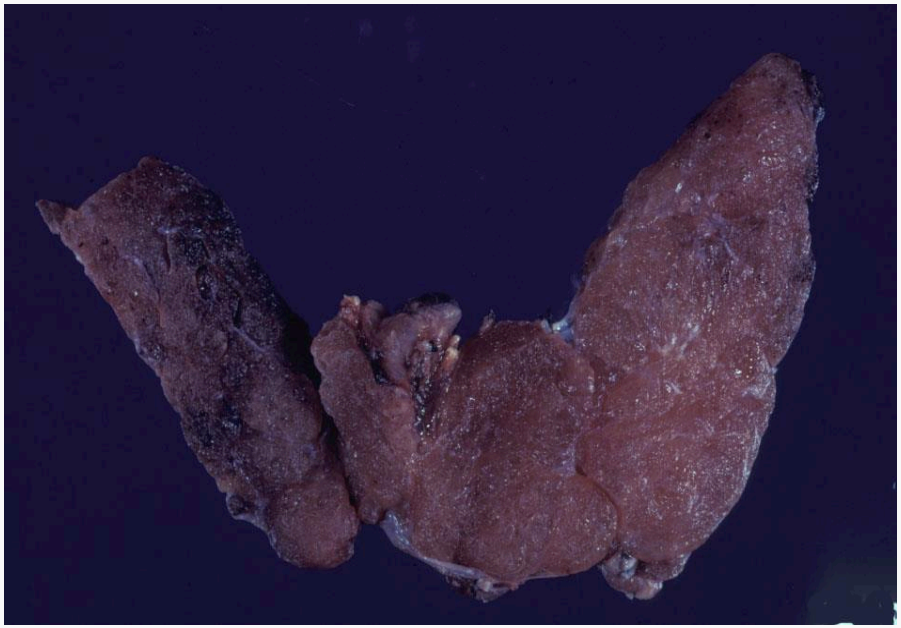


Pretibial myxedema

Graves’ disease (diffuse hyperplasia)



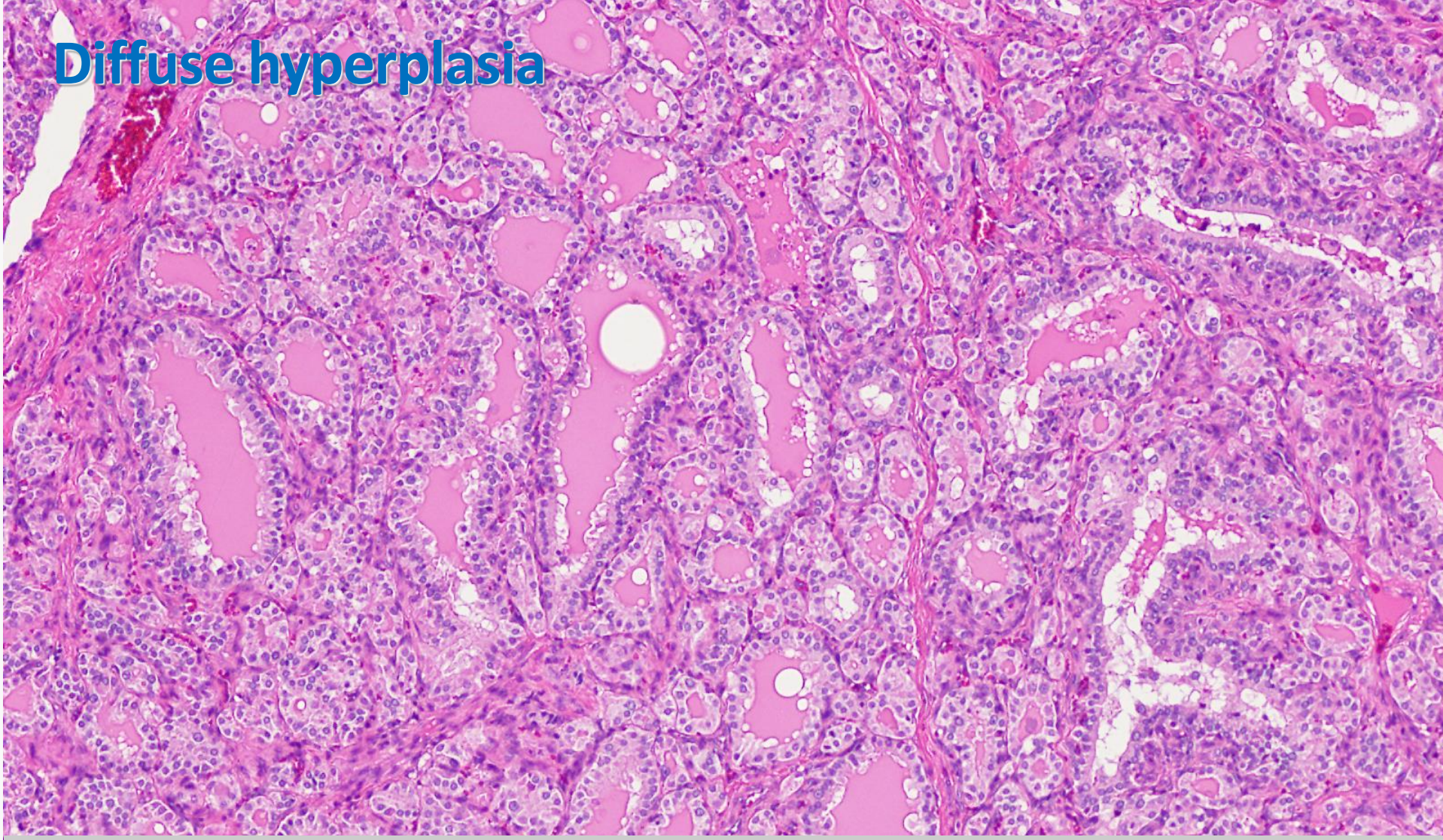
Fresh specimen



Formalin-fixed specimen

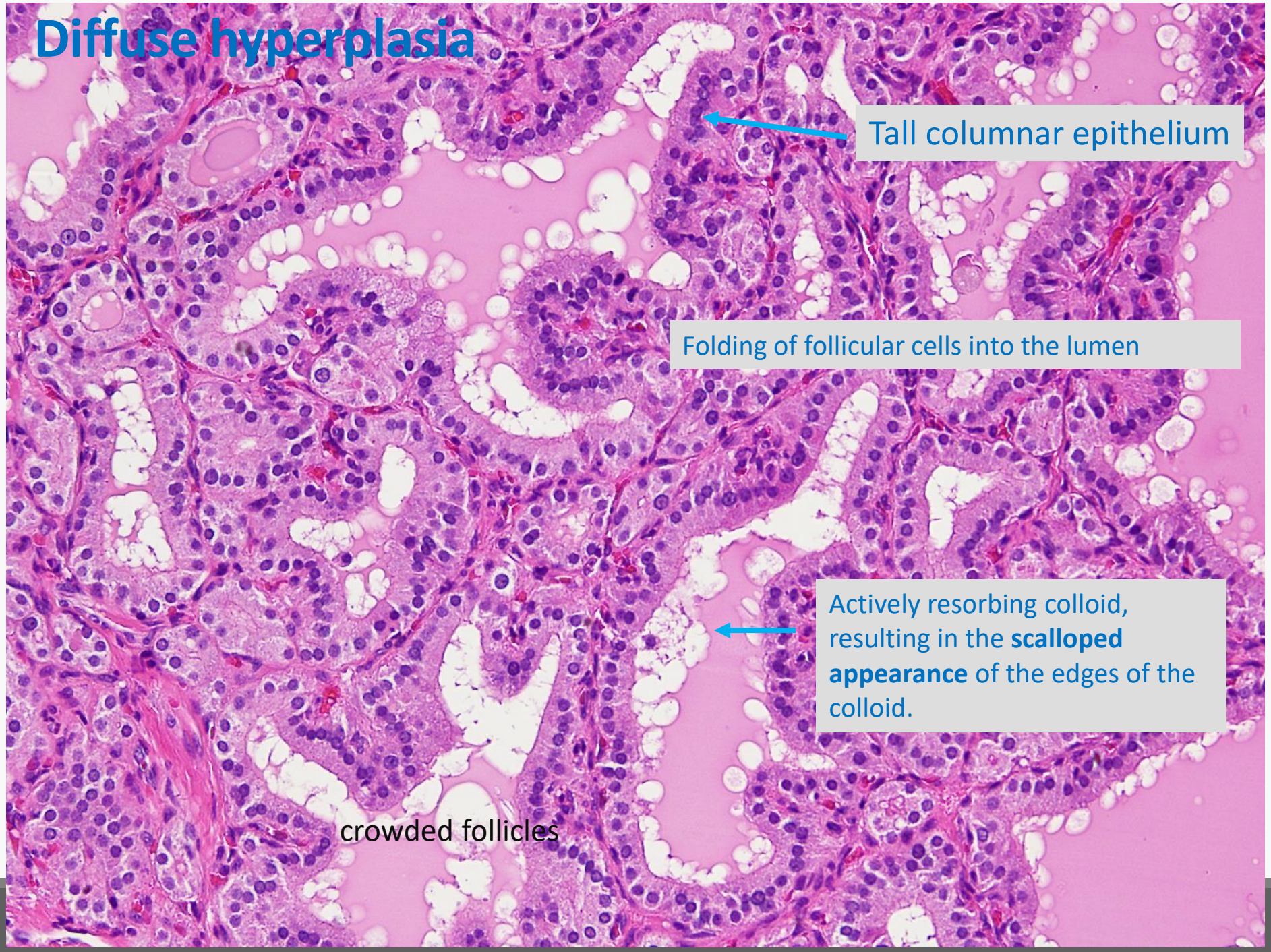
Diffuse, symmetrical enlargement of both lobes

Diffuse hyperplasia



- crowded follicles
- folding of epithelial cells into the lumens.
- **tall columnar** shape of follicular cells.
- **scalloped** appearance of the edges of the colloid.

Diffuse hyperplasia



Thyroiditis

- Infectious thyroiditis
- Subacute granulomatous thyroiditis
- Hashimoto’s thyroiditis
- Silent Lymphocytic thyroiditis
- Fibrous [Reidel] thyroiditis)

Infectious thyroiditis

Infectious thyroiditis may be either acute or chronic.

Acute infections via **hematogenous spread** or through direct seeding of the **adjacent** structure

Other infections of the thyroid, eg mycobacterial, fungal frequently occur in immunocompromised patients

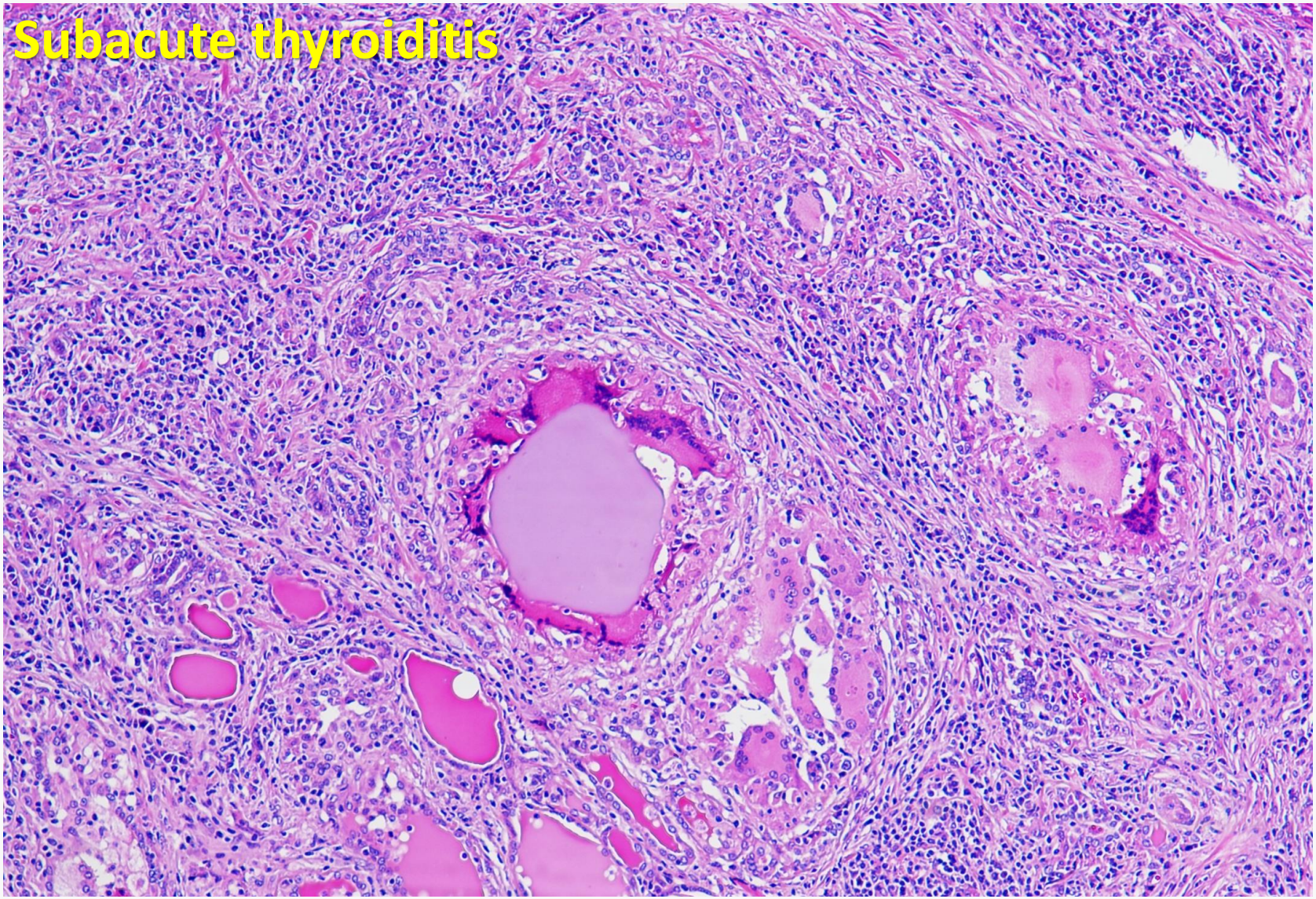
Subacute thyroiditis

Other names: De Quervain's thyroiditis, giant cell thyroiditis subacute granulomatous thyroiditis

Clinical

- sudden, painful enlargement of the thyroid gland.
- Often affect young and middle-aged women.
- Usually exhibits evidence of hyperthyroidism, but Hypothyroidism is not uncommon.
- Generally, it lasts a few weeks or months.

Subacute thyroiditis



Inflammatory infiltrate composed of multinucleated giant cells, foamy histiocytes, epithelioid histiocytes, neutrophils, lymphocytes, plasma cells

Subacute thyroiditis



Multinucleated giant cells engulfing colloid

Hashimoto's thyroiditis

Chronic autoimmune disease

Female: male 3-12:1

Age: 30-50 years

- **Pathogenesis**

  : Autoantibodies against TSH receptor, microsome, thyroglobulin

  : Follicular cell destruction by cytotoxic T-cell

Hashimoto's thyroiditis

Clinical features

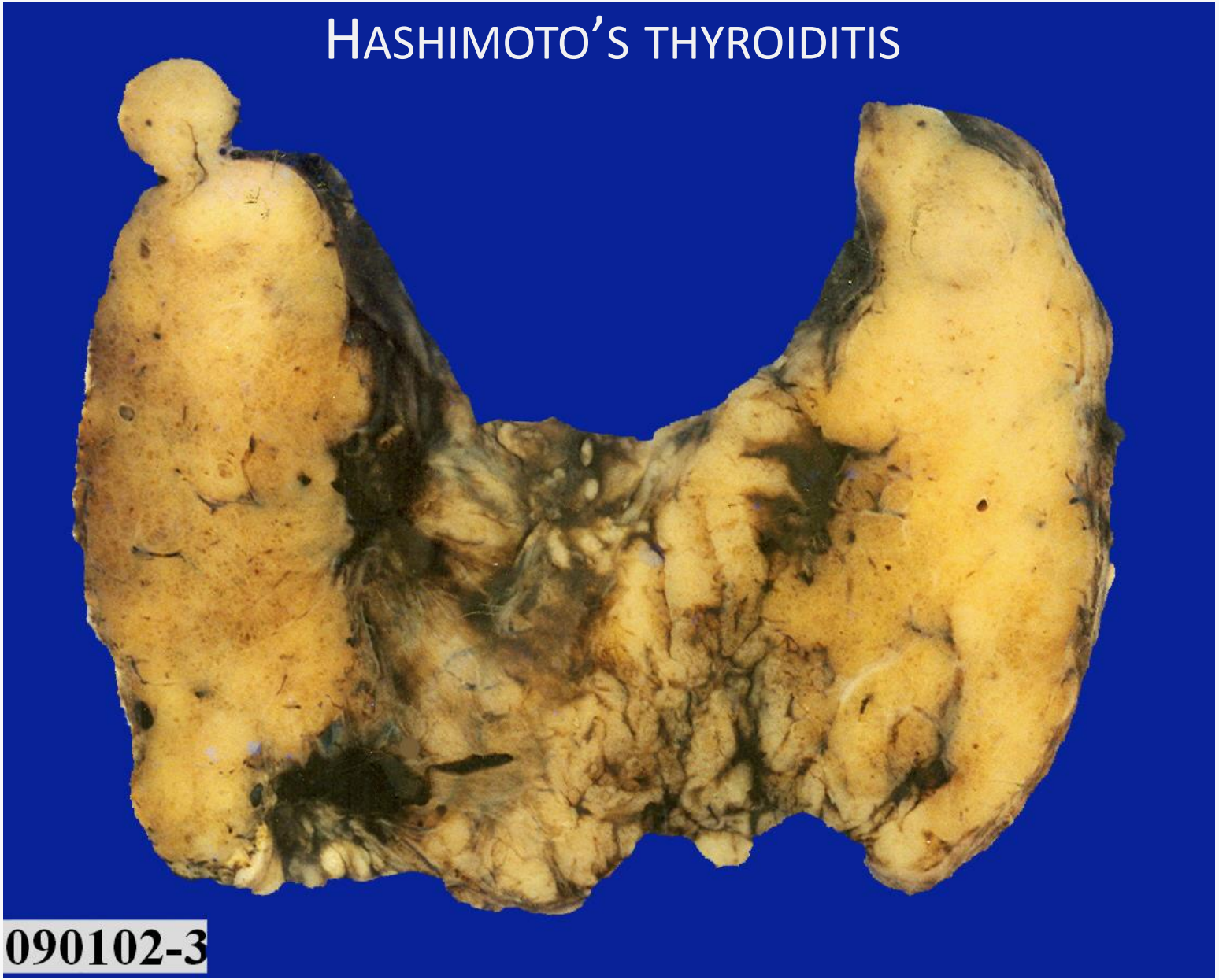
- Euthyroid or transient hyperthyroid ->> hypothyroid
- may coexist with other autoimmune diseases eg. SLE, rheumatoid arthritis, chronic active hepatitis, etc
- Increased risk of primary thyroid lymphoma and papillary thyroid carcinoma



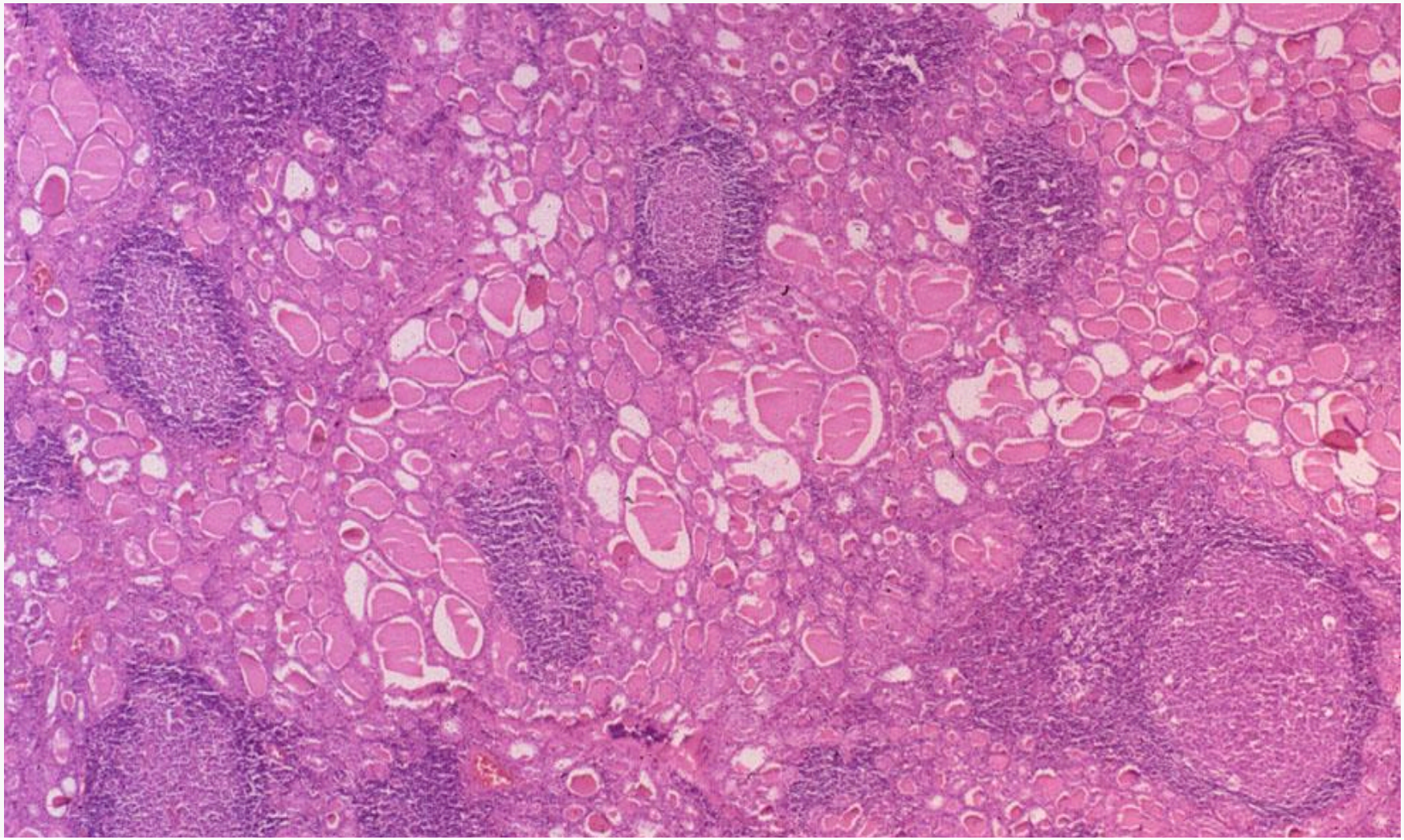
Gross: Hashimoto's thyroiditis

- Diffuse symmetric enlargement of thyroid gland
- Occasionally, the gland is nodular or asymmetric or atrophy
- Cut surface resembles lymph nodes with tannish yellow or gray-white color

HASHIMOTO'S THYROIDITIS

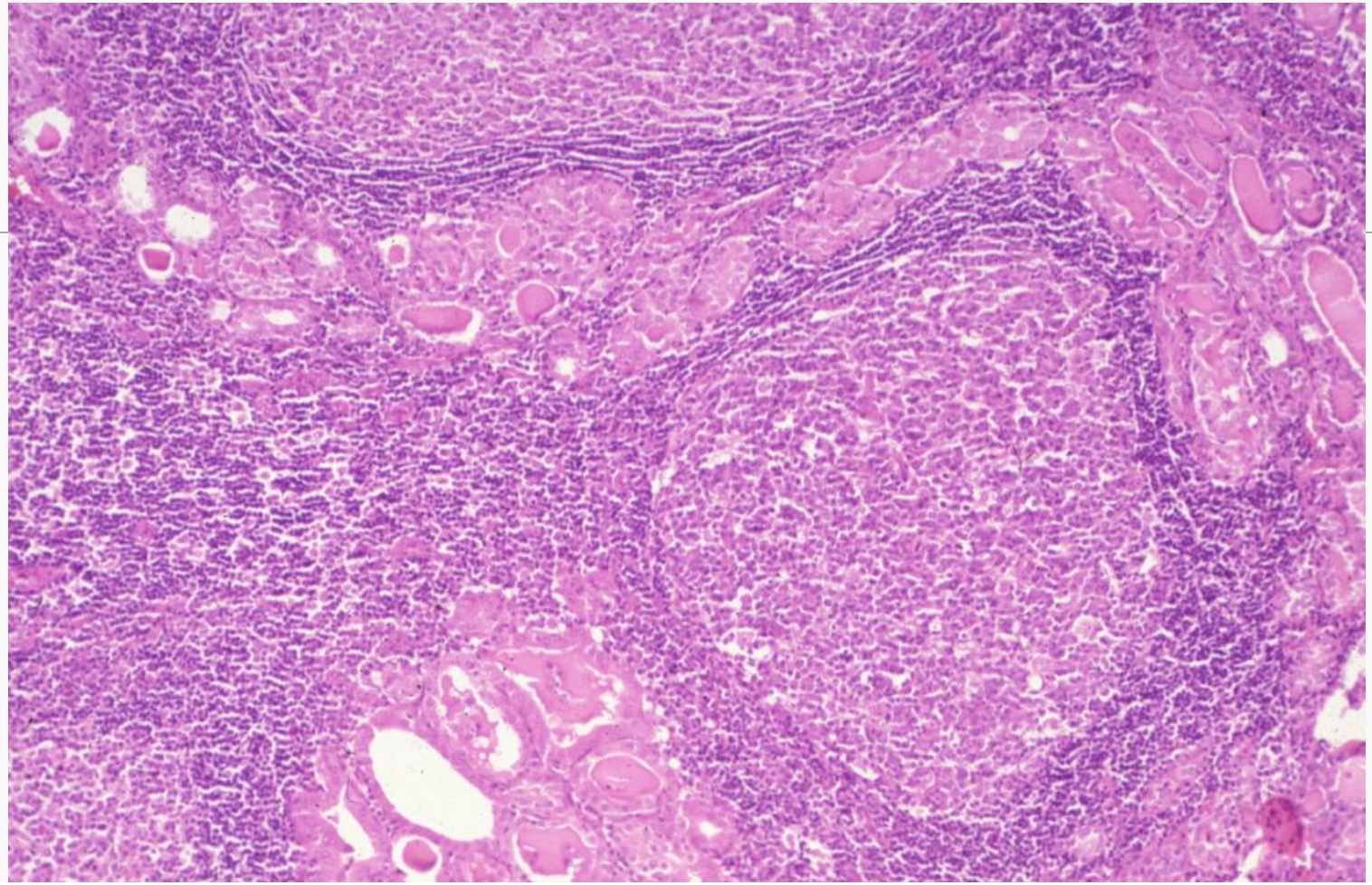


Hashimoto's thyroiditis



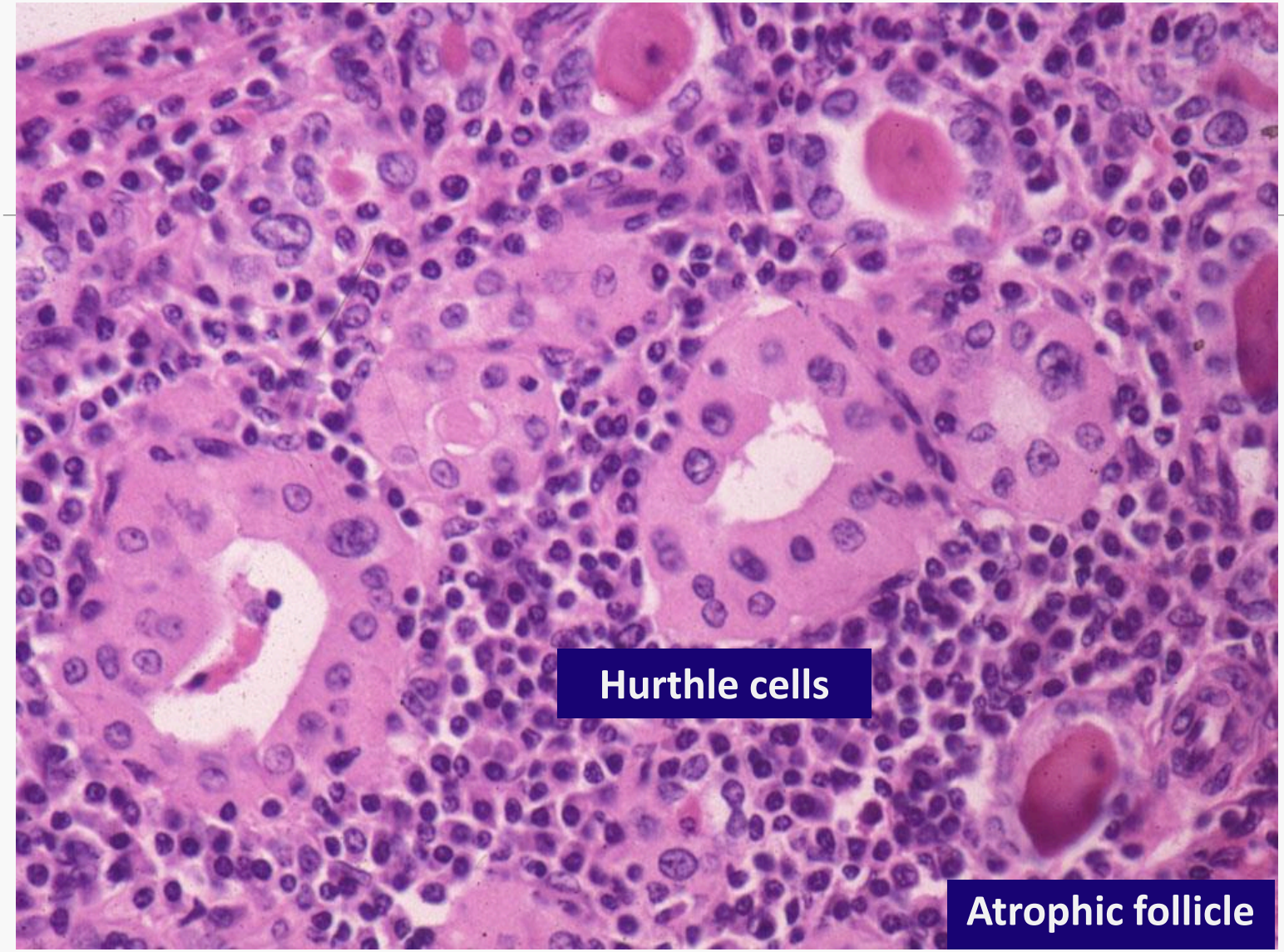
- Infiltration of lymphocytes and plasma cells
- lymphoid follicles with germinal center formation
- Atrophic follicles with Hürthle cells (**follicular cells with abundant eosinophilic cytoplasm**)

Hashimoto's thyroiditis



- Infiltration of lymphocytes and plasma cells
- lymphoid follicles with germinal center formation
- Atrophic follicles with Hürthle cells (**follicular cells with abundant eosinophilic cytoplasm**)

Hashimoto's thyroiditis



Silent lymphocytic thyroiditis

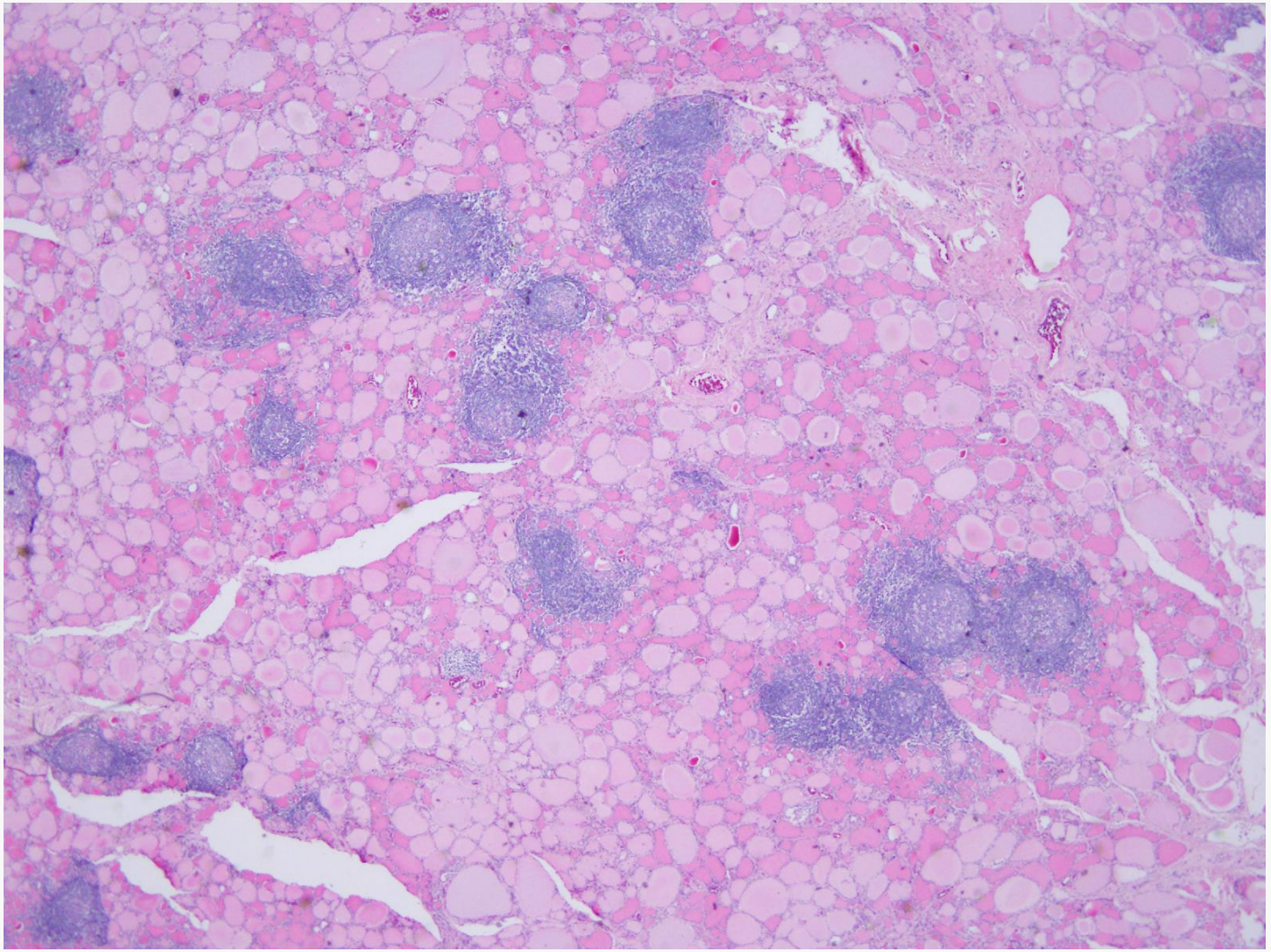
The pathogenesis: unknown

Histology: Many lymphocytes, no Hurthle cells

Symptom:

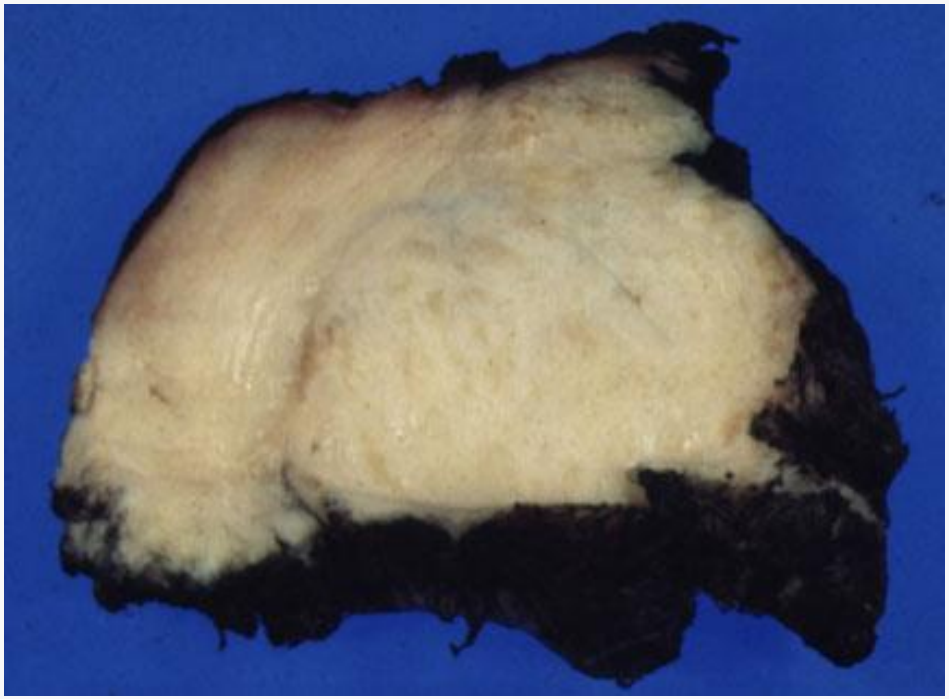
- asymptomatic
- or present with a painless, slightly enlarged thyroid.
- mild, may has transient hyperthyroidism

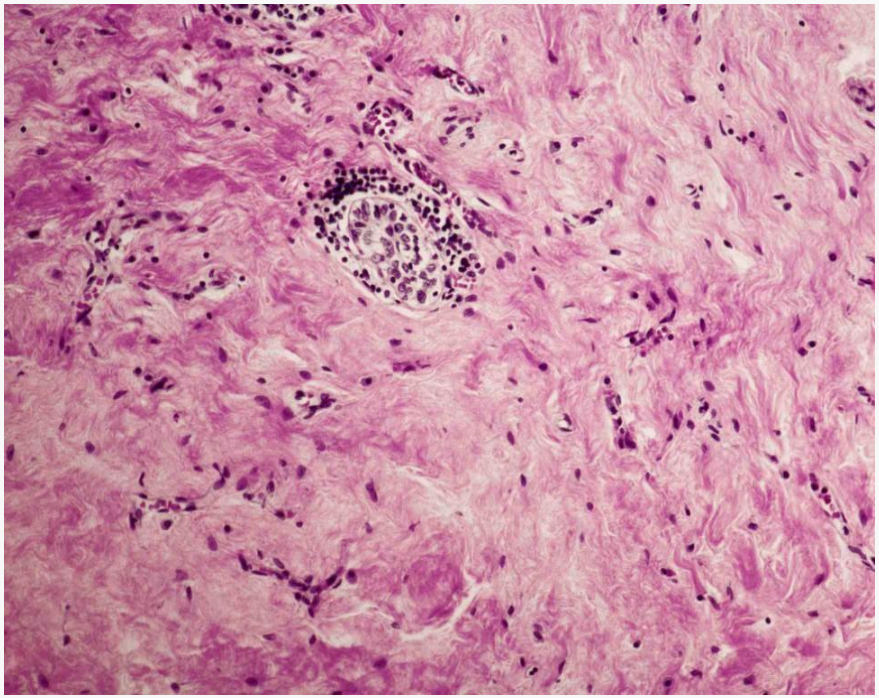
Silent lymphocytic thyroiditis



Fibrous (Reidel) thyroiditis

- Unknown etiology - could be a variant of autoimmune thyroiditis or part of a systemic fibrosing disorder.
- thyroid parenchyma and surrounding structures are replaced with dense fibrous tissue





Multinodular goiter

Other names: non-toxic goiter, endemic goiter

Updated name: Thyroid follicular nodular disease (FND)

Etiology: ขาด iodine, chemical, inborn error

Age: variables

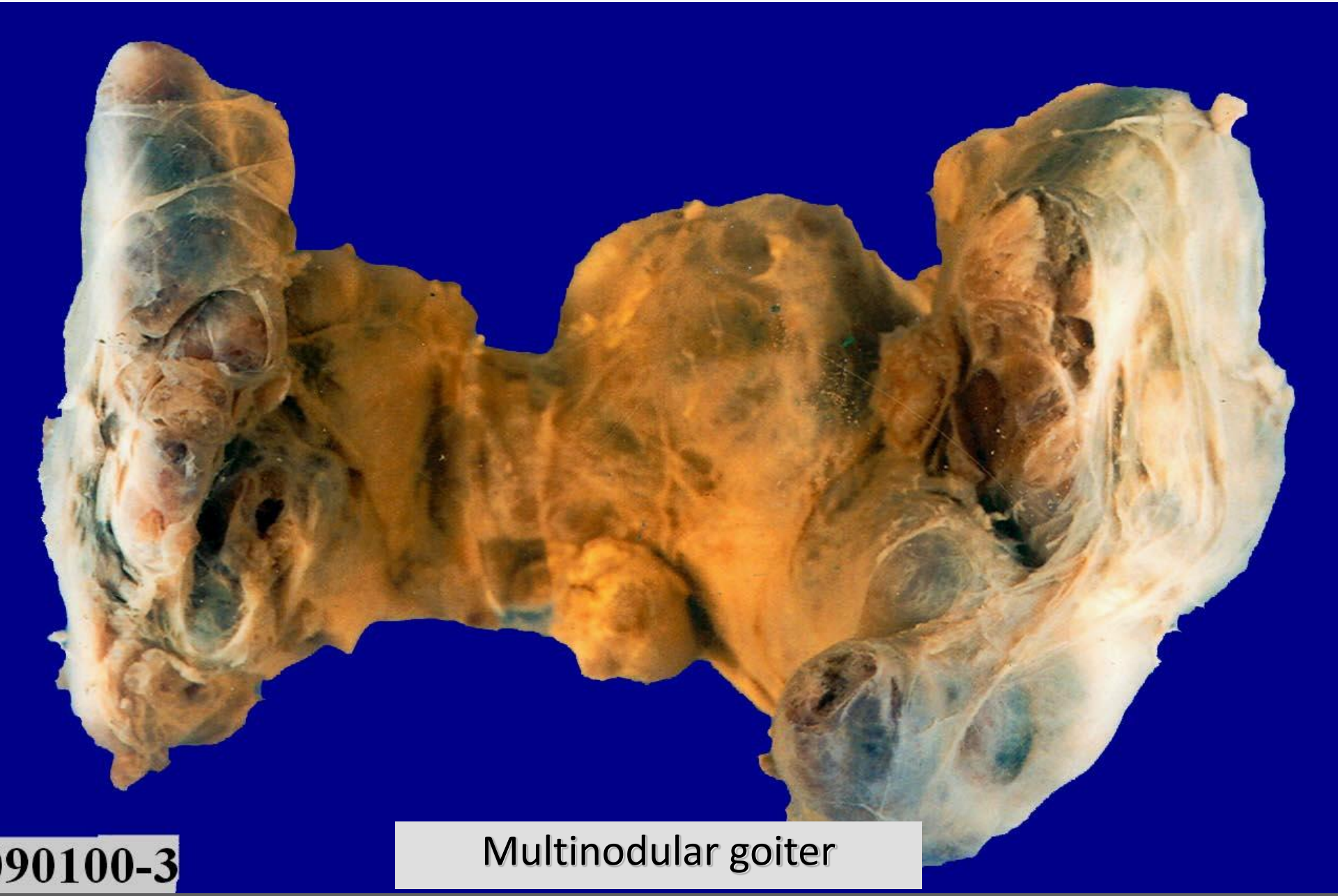
Female >>> Male

Clinical: Thyroid enlargement, nodularity

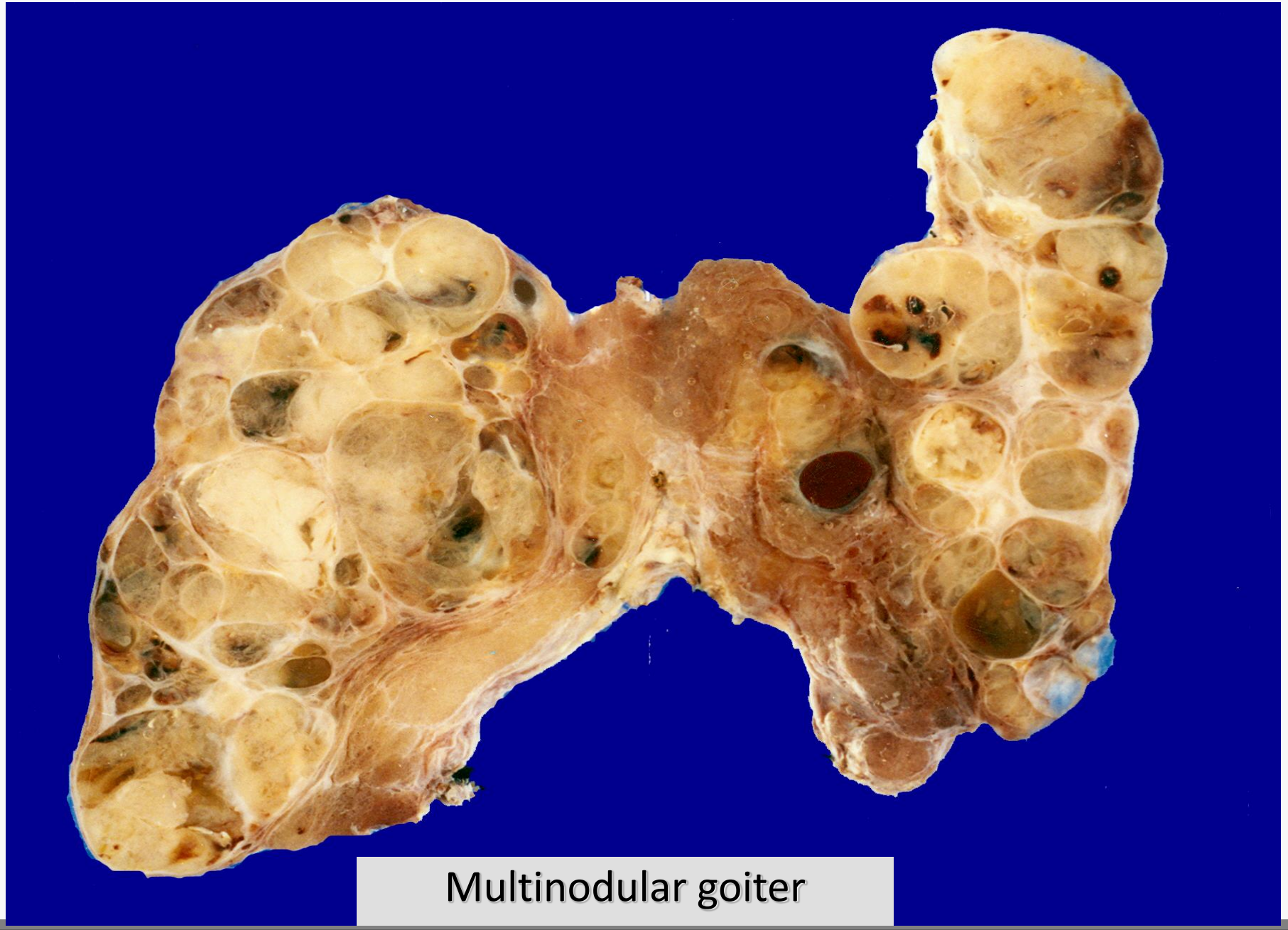
: Euthyroid (no hyper or hypothyroidism)

: Pressure effect- rare

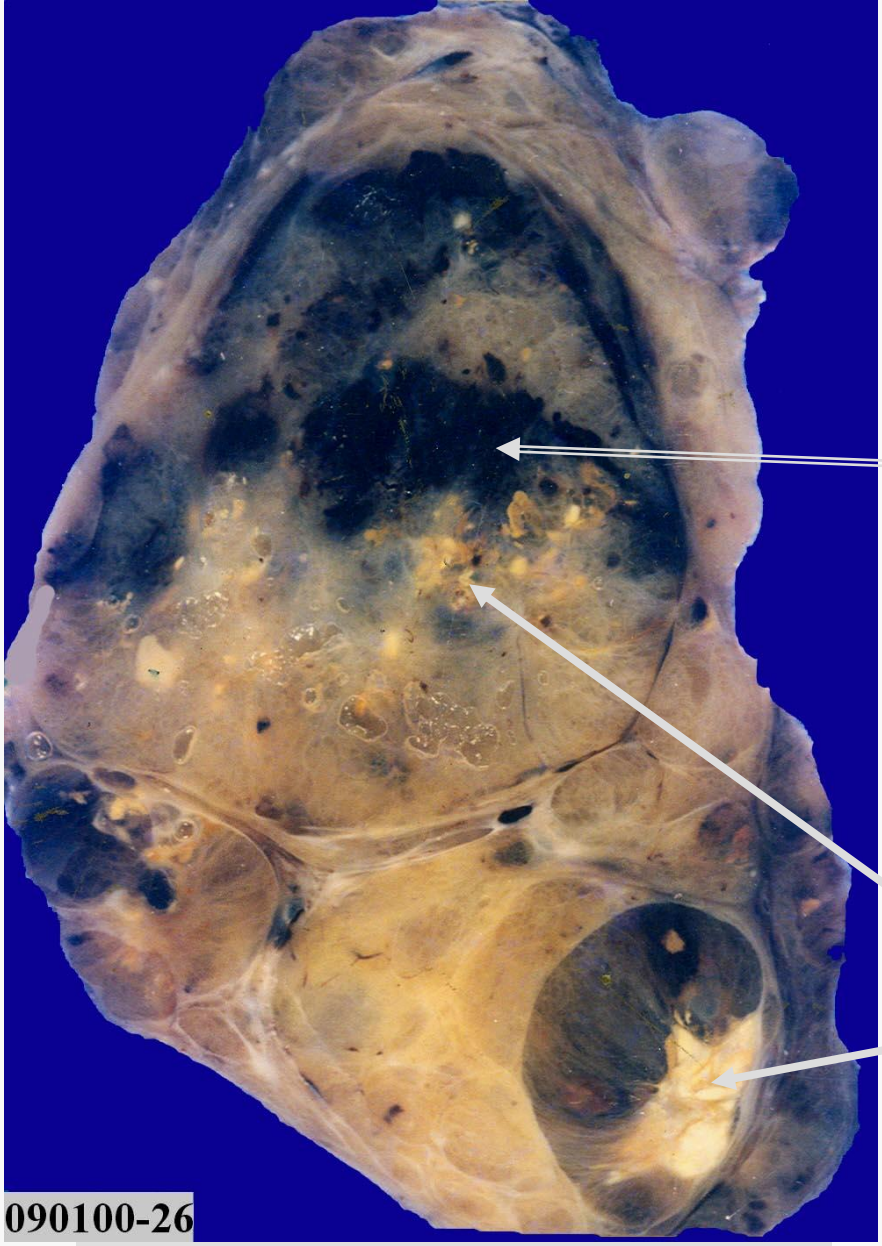

Nodular goiter



Multinodular goiter



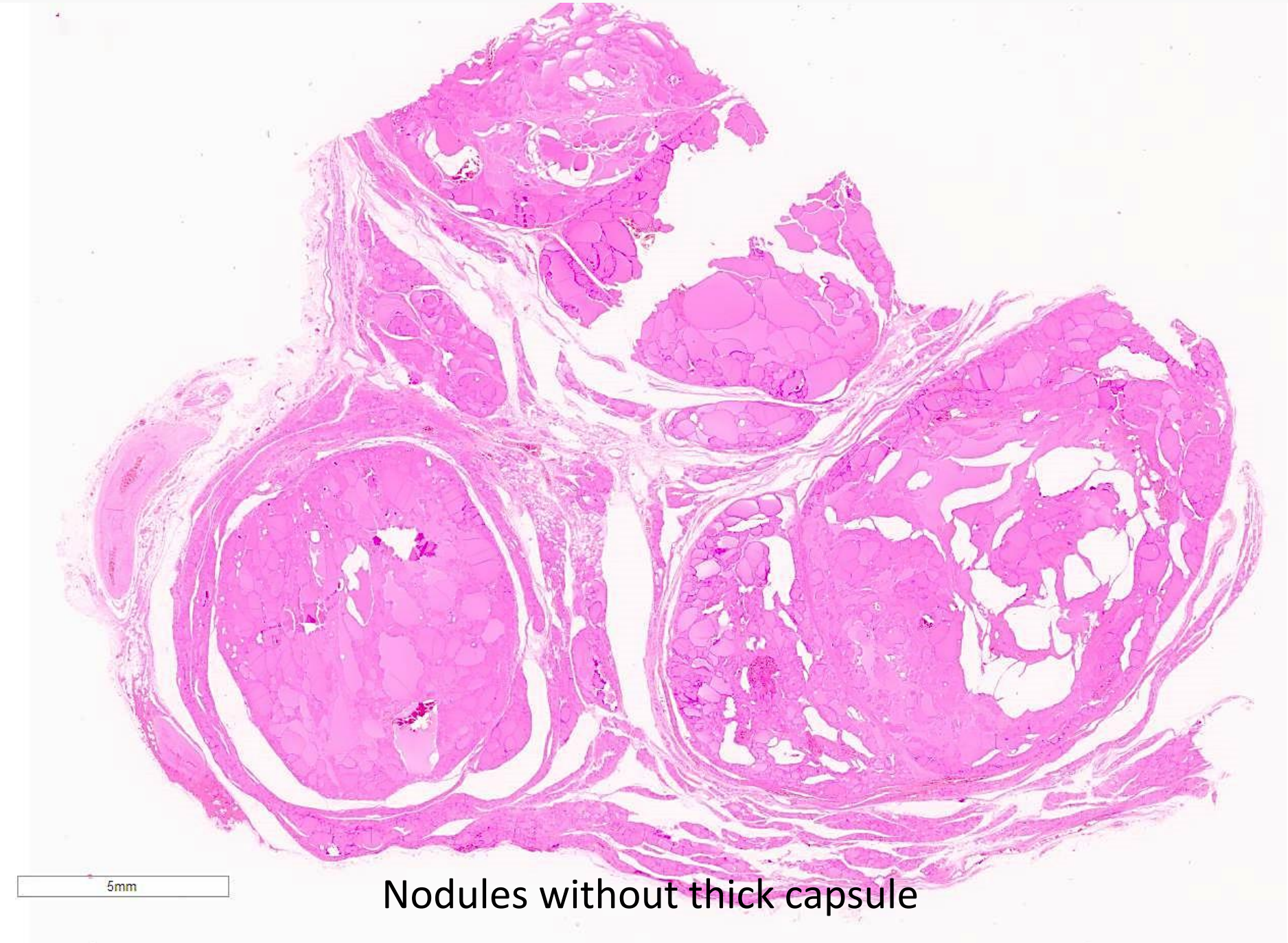
Multinodular goiter



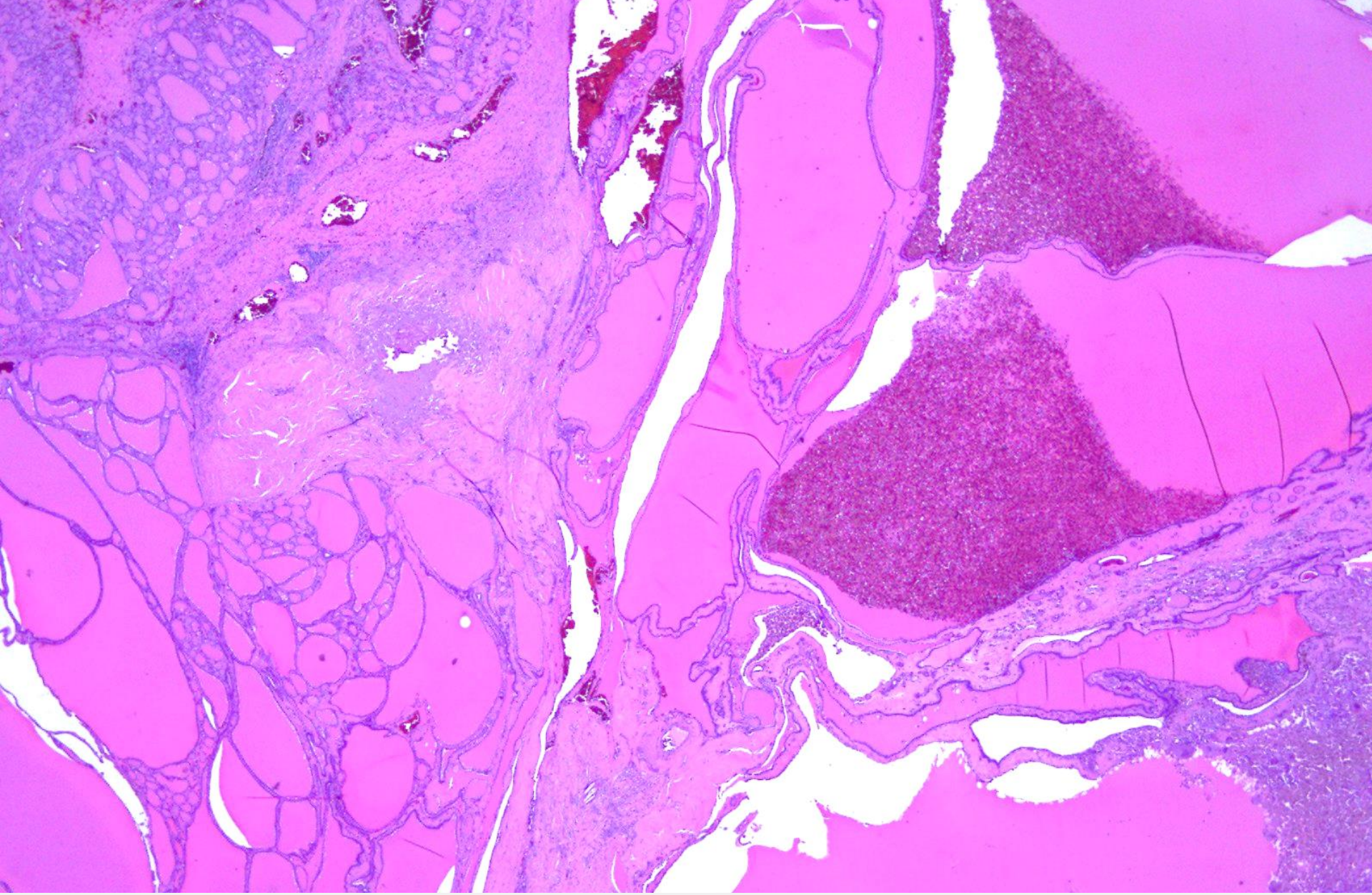
Multinodular goiter



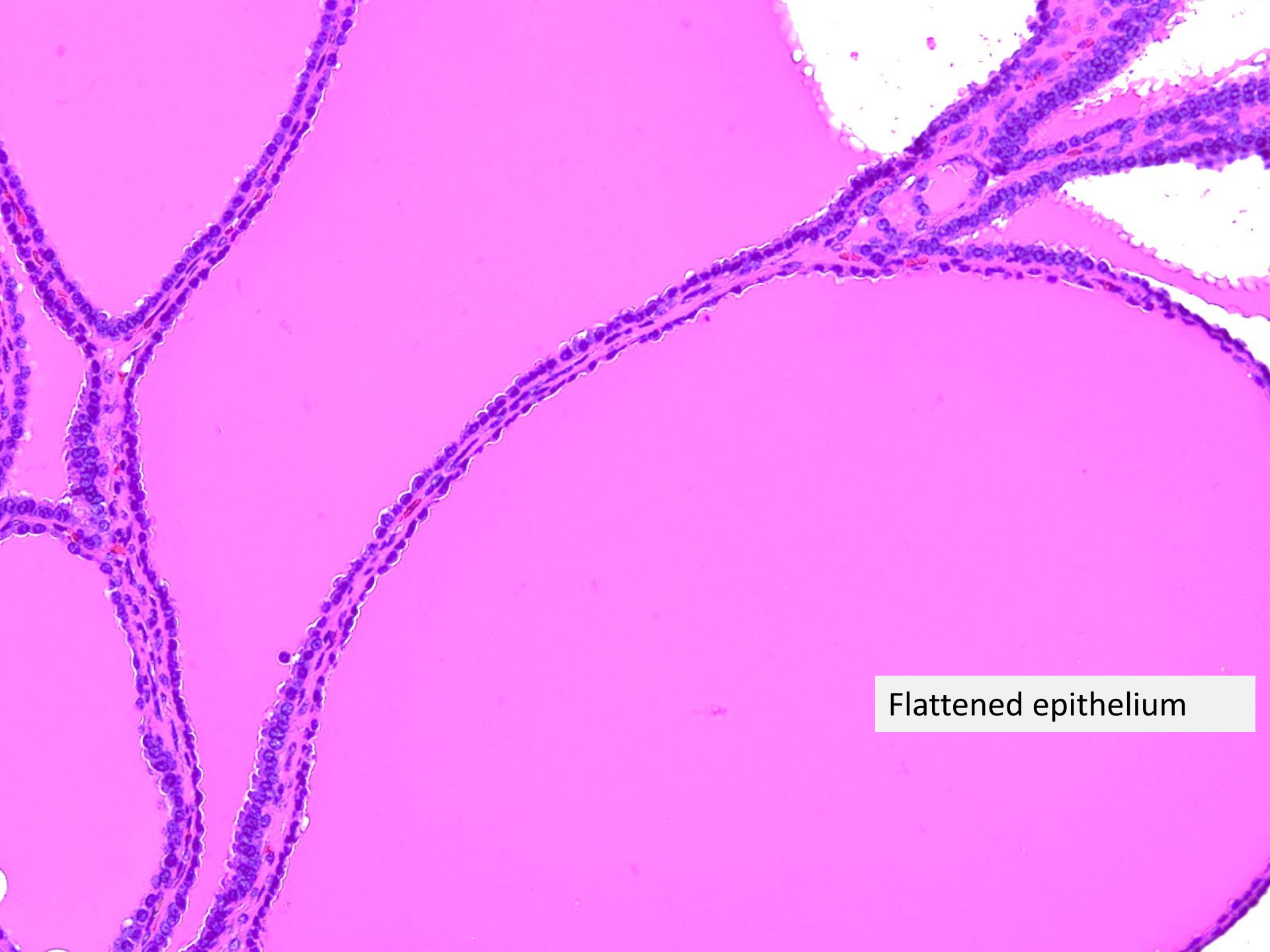
Multinodular goiter



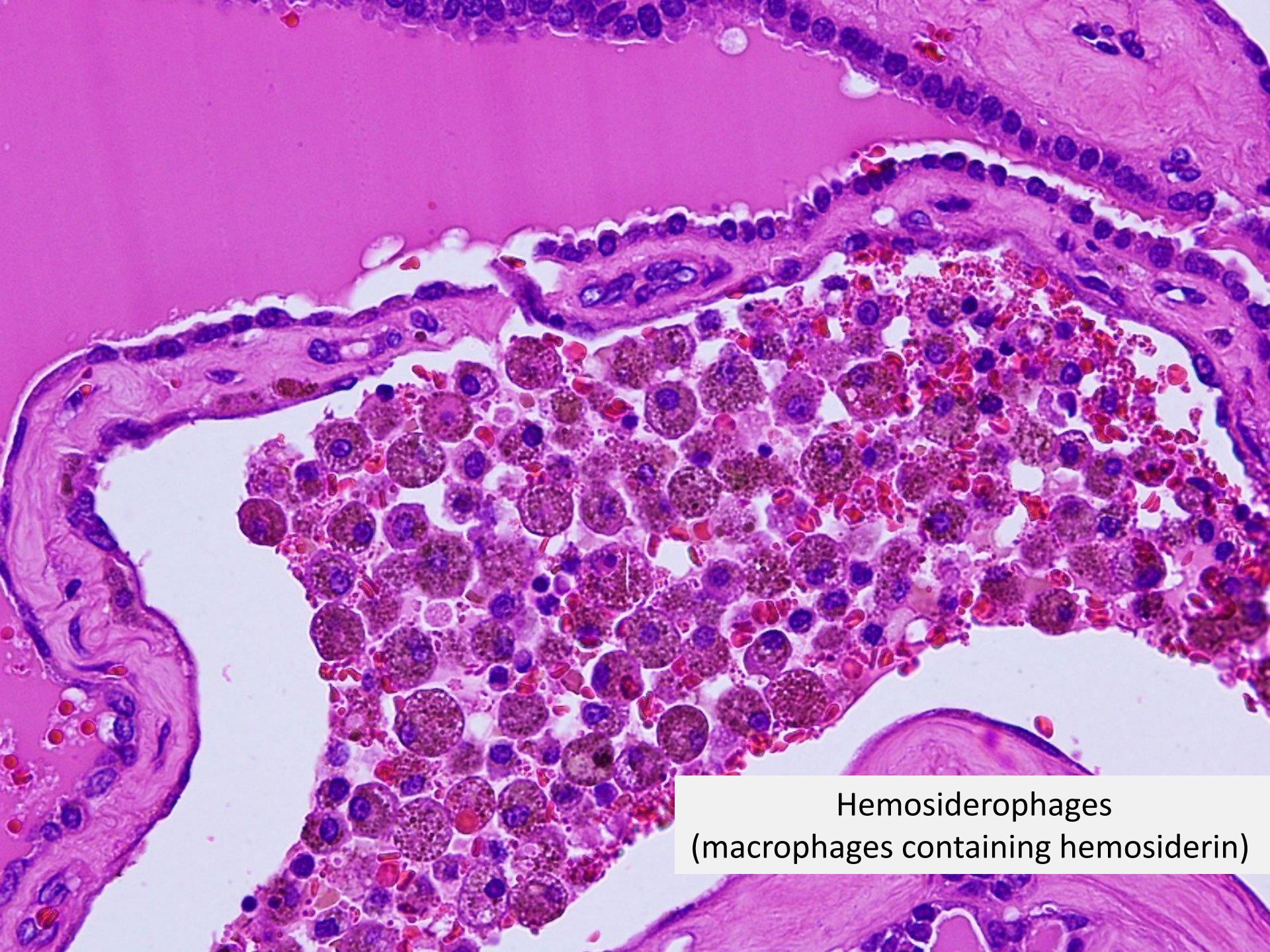
Nodules without thick capsule



Variable-sized dilated follicles with flattened to hyperplastic epithelium


Flattened epithelium


Hemorrhage
Hemosiderophages
(macrophages containing hemosiderin)


Hemosiderophages
(macrophages containing hemosiderin)


Fibrosis and calcification

- Toxic multinodular goiter (Plummer's disease)

  Older patients with prolonged nodular goiter

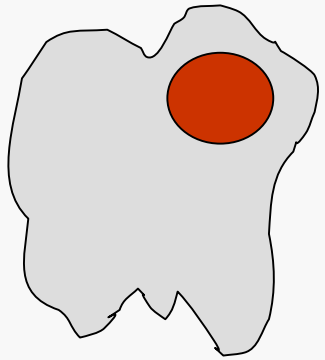
- Simple goiter (diffuse non-toxic goiter)

  -> early phase of nodular goiter

  -> adolescent, pregnancy

Solitary thyroid nodule

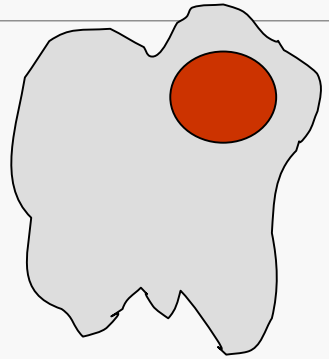
- A disproportionately prominent nodule in nodular goiter
- Benign tumor
- Malignant tumor



Tumor of thyroid gland

- Benign

  Follicular adenoma

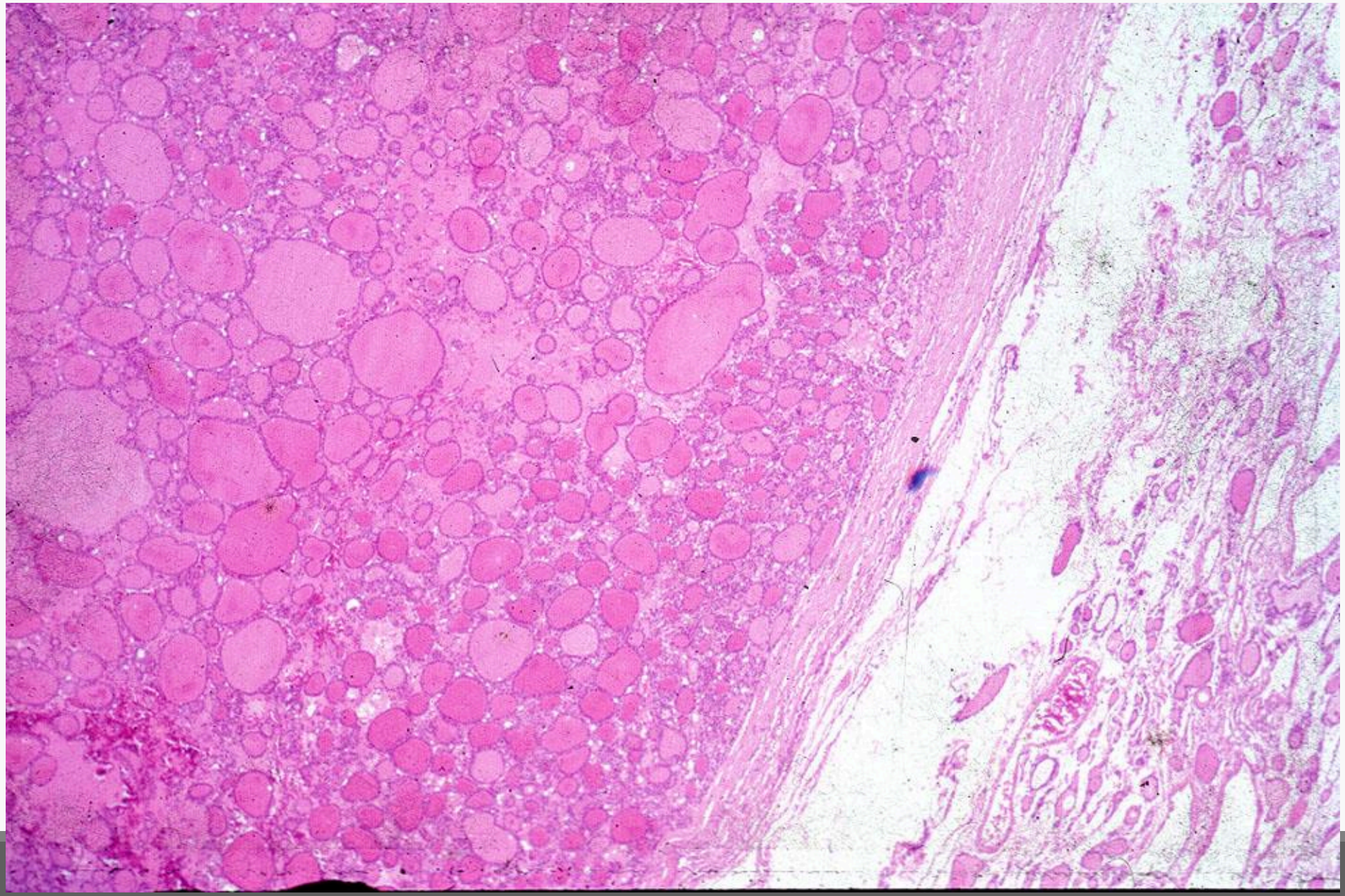


- Malignant tumor
  - Papillary carcinoma
  - Follicular carcinoma
  - Undifferentiated carcinoma
  - Medullary carcinoma
  - Malignant lymphoma

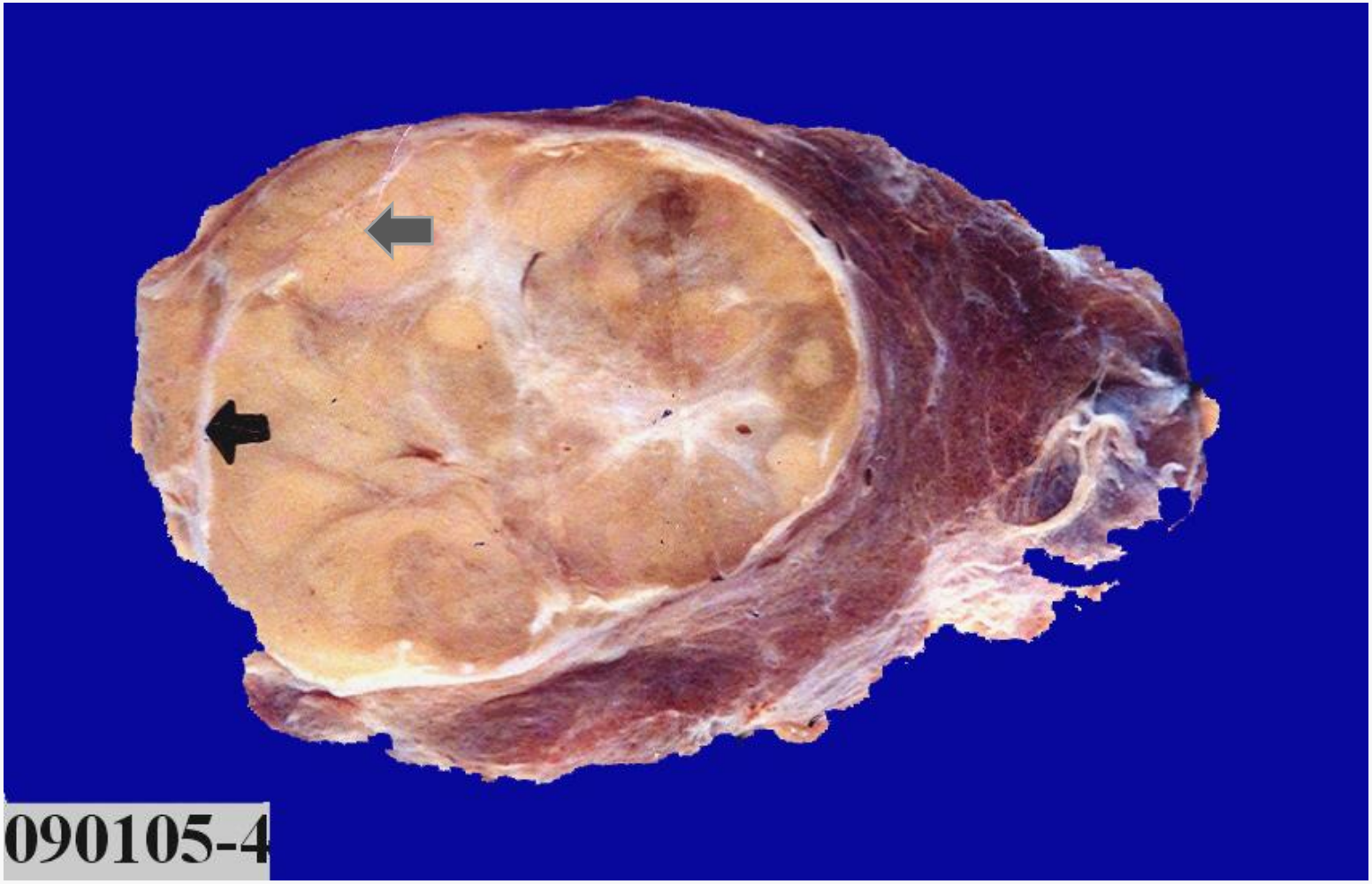
Follicular adenoma



Follicular adenoma



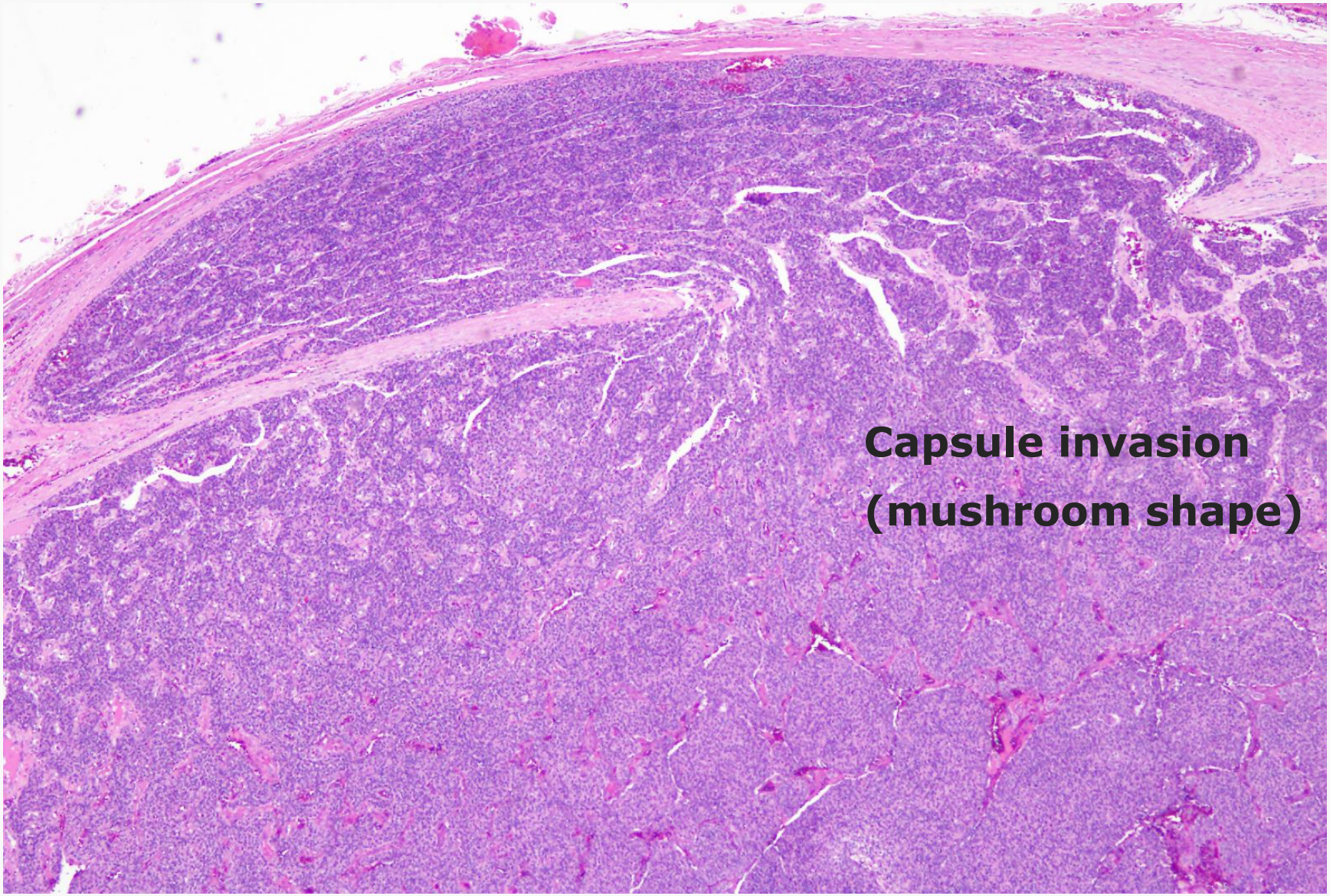
Follicular carcinoma



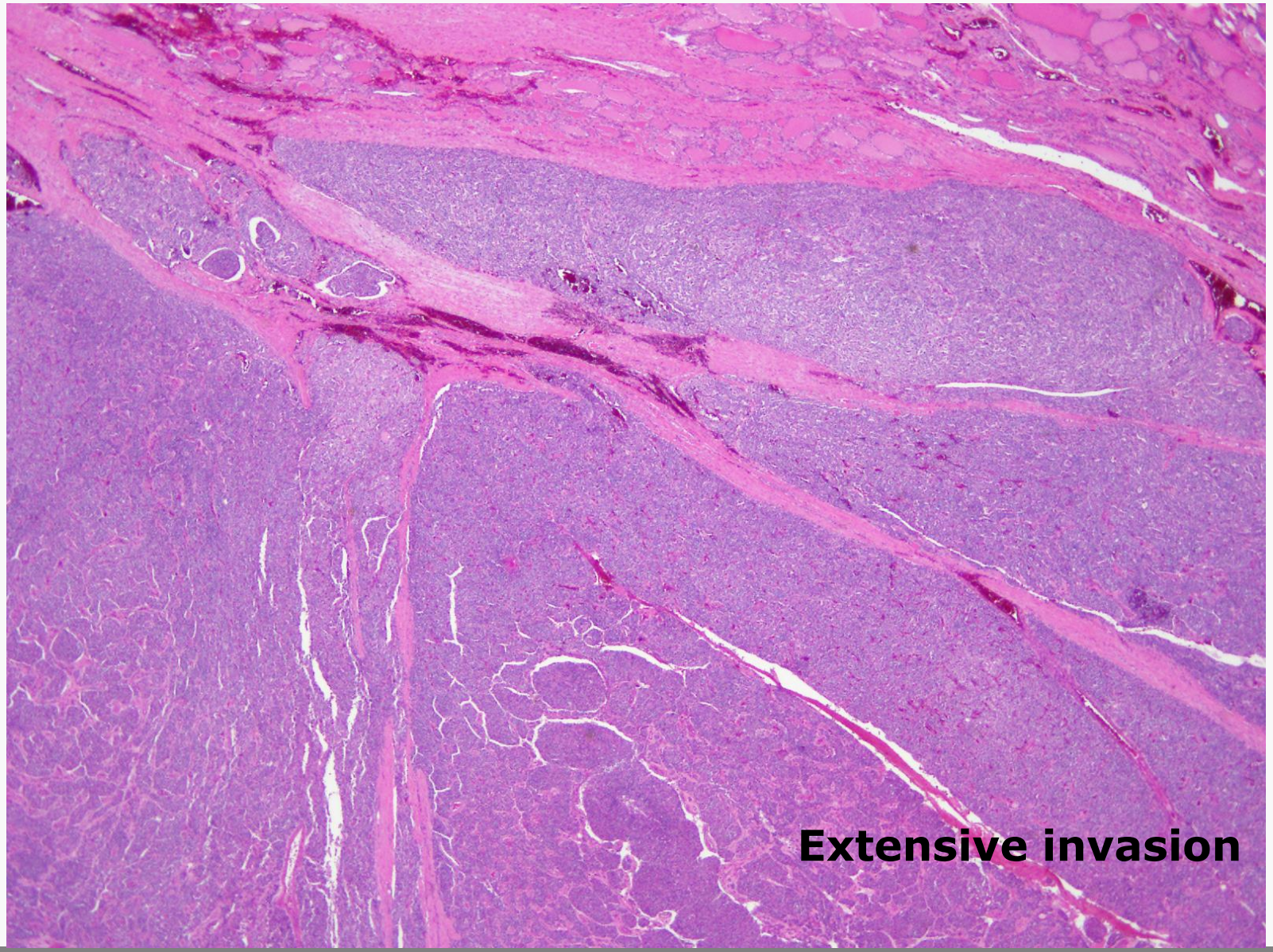
Follicular carcinoma



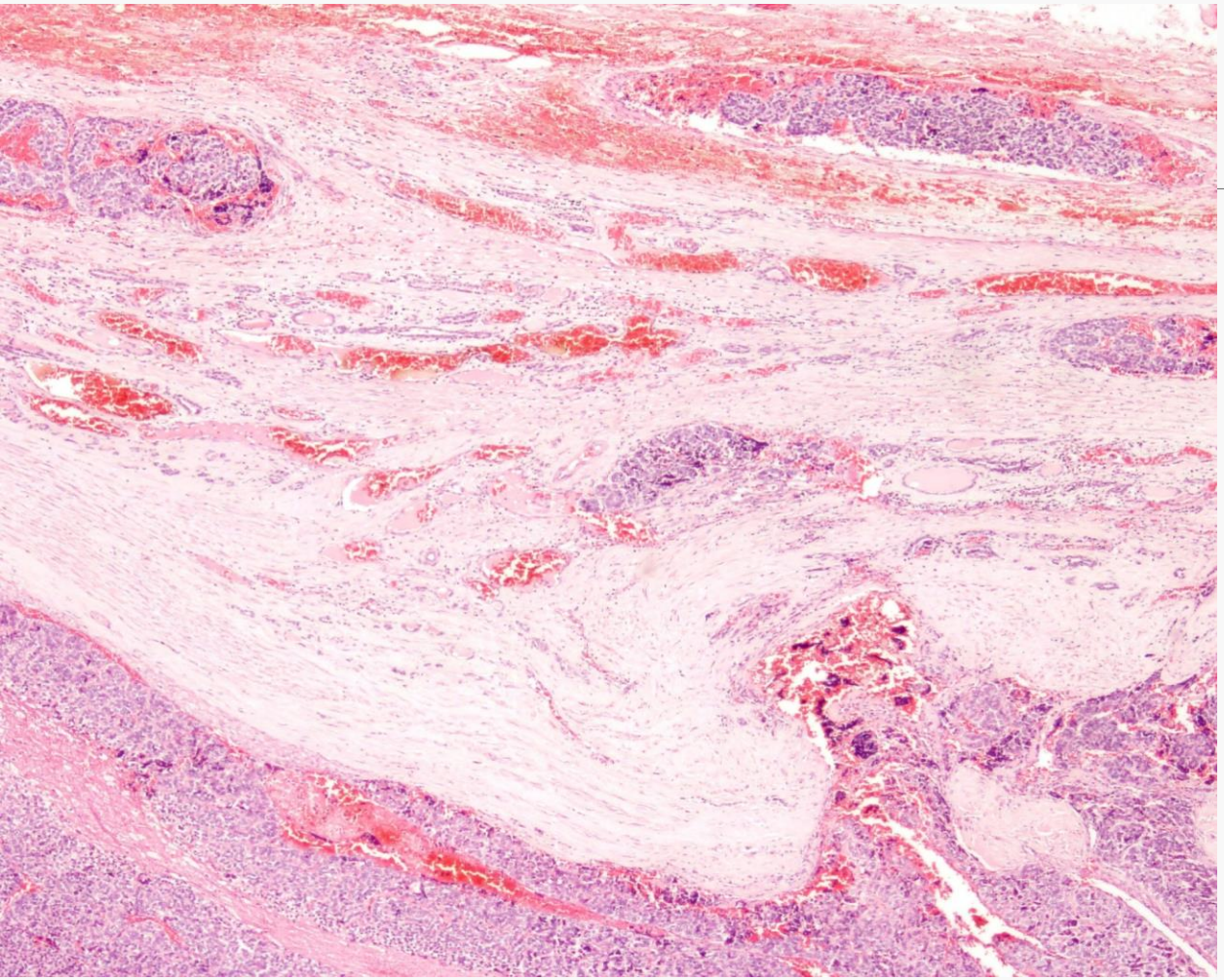
Follicular carcinoma



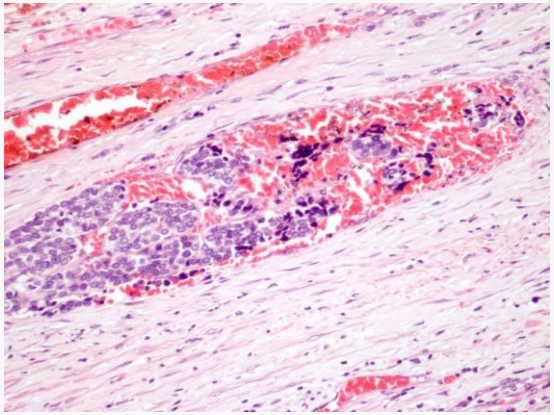
Follicular carcinoma



Follicular carcinoma



Vascular invasion



Papillary Carcinoma

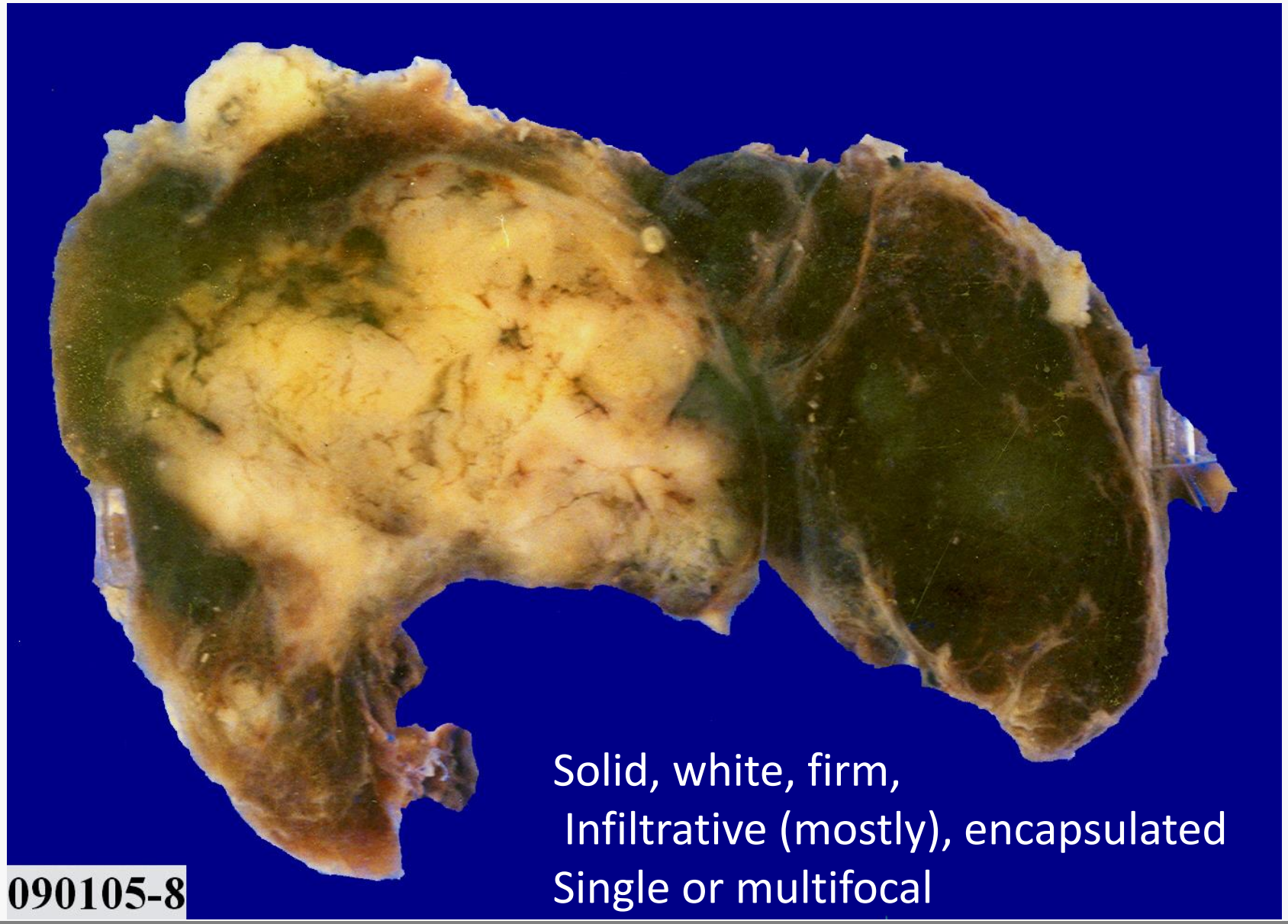
- Most common (70%) thyroid cancer
- พบได้ทุกอายุ
- Etiology: Ionizing radiation before age 20,

  Hashimoto thyroiditis, possibly
- การแพร่กระจายทางน้ำเหลือง
- พยากรณ์โรคดีมาก

  อัตรารอดชีพ 20 ปี > 90%

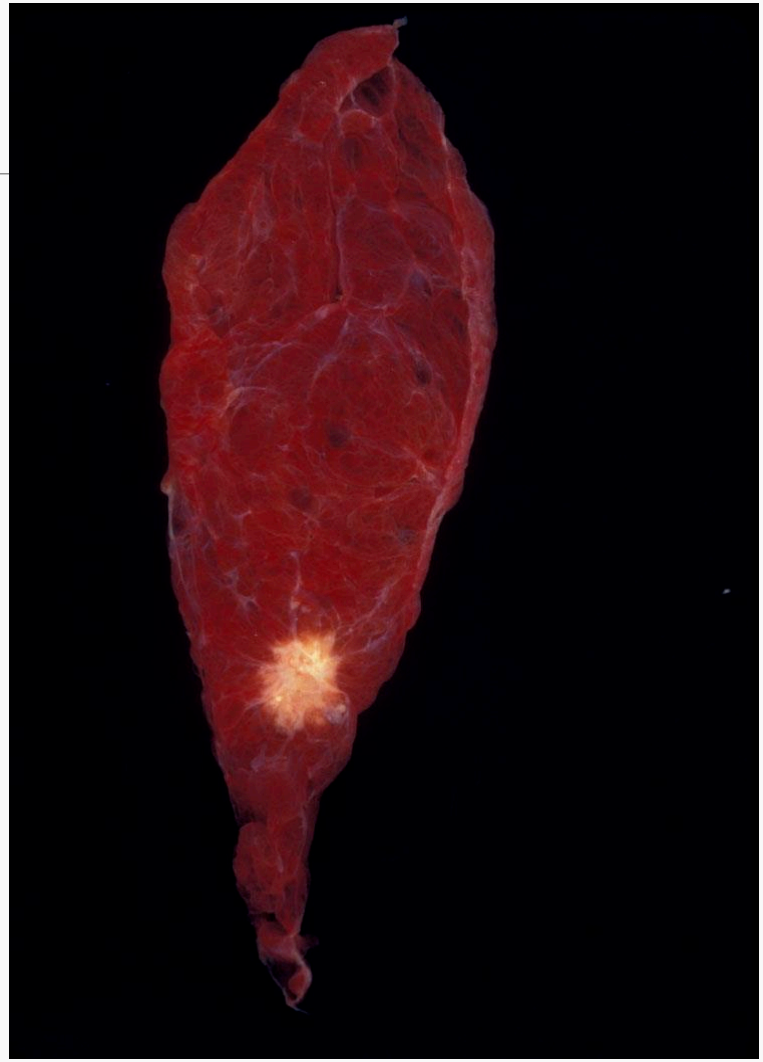
  อัตรารอดชีพในคนอายุน้อยเกือบ 100%

Papillary carcinoma

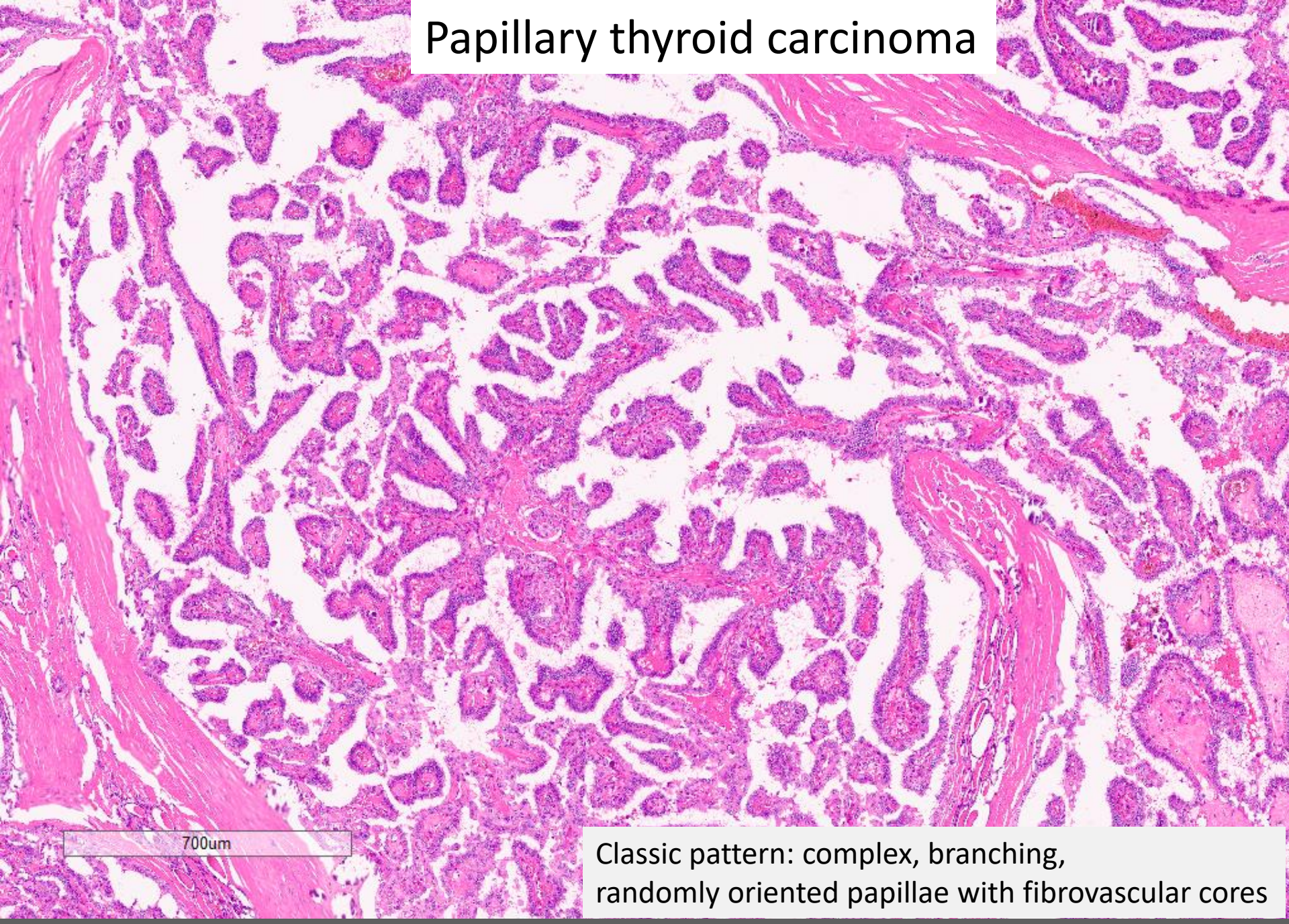


Papillary Carcinoma



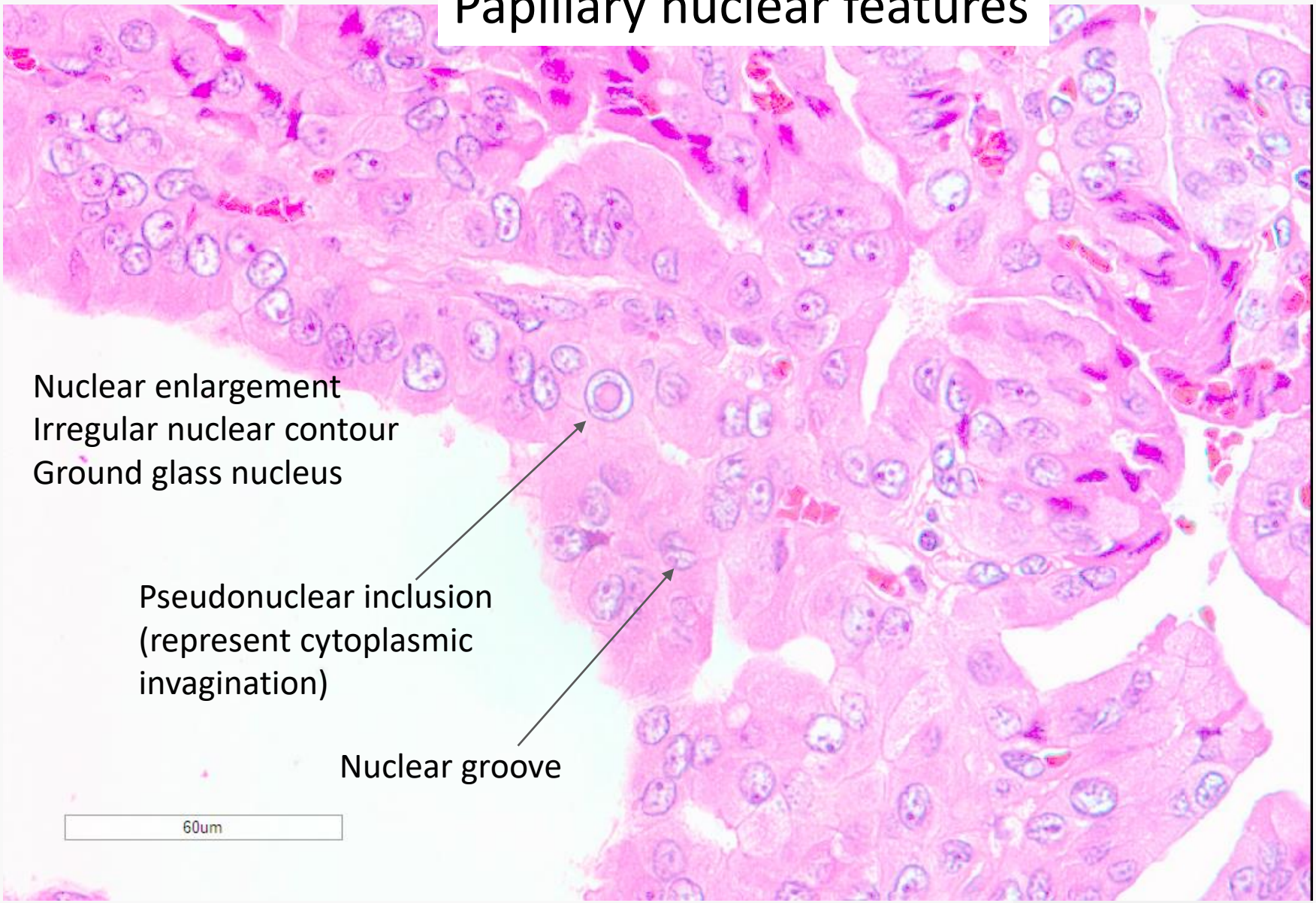


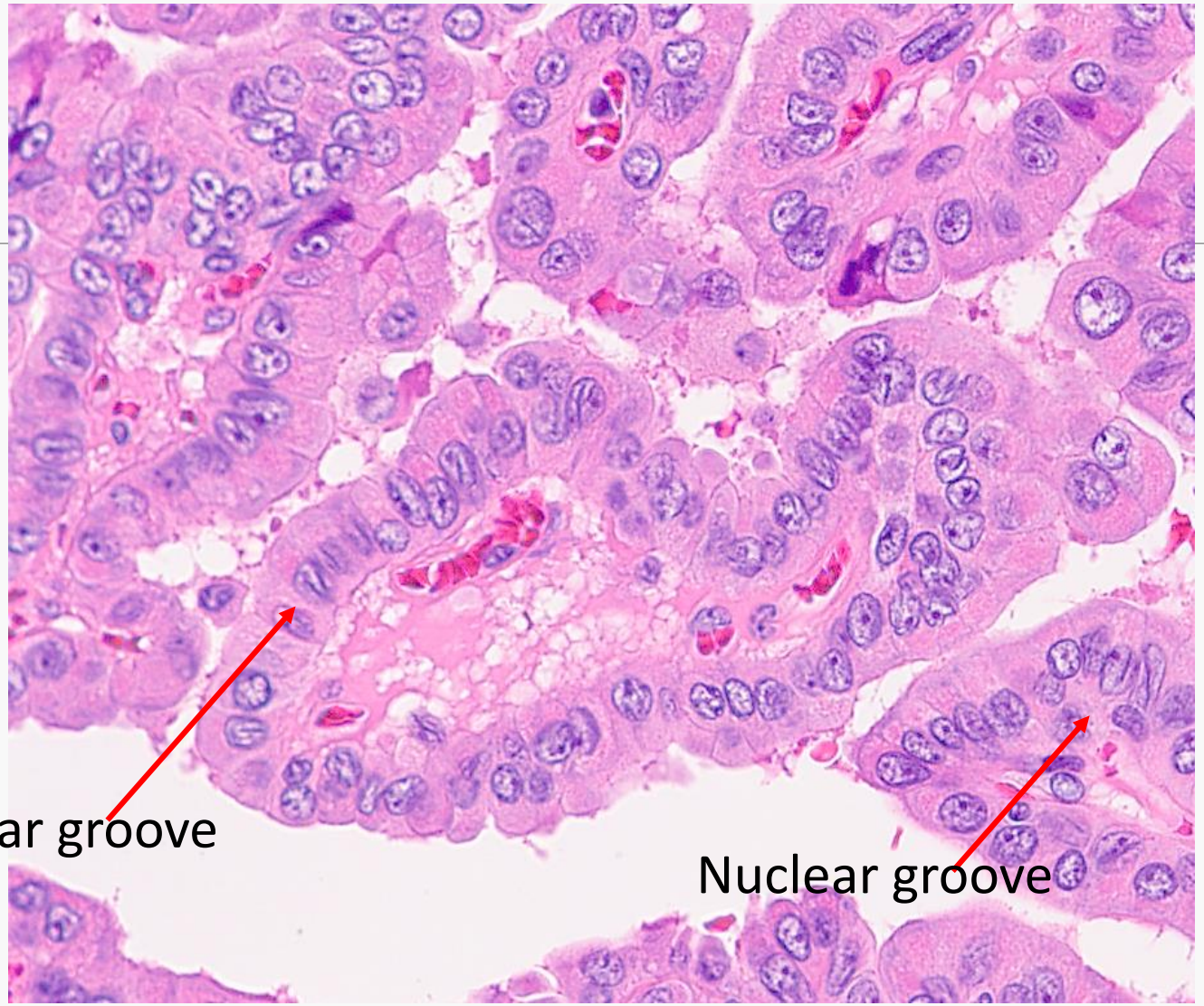
Papillary thyroid carcinoma



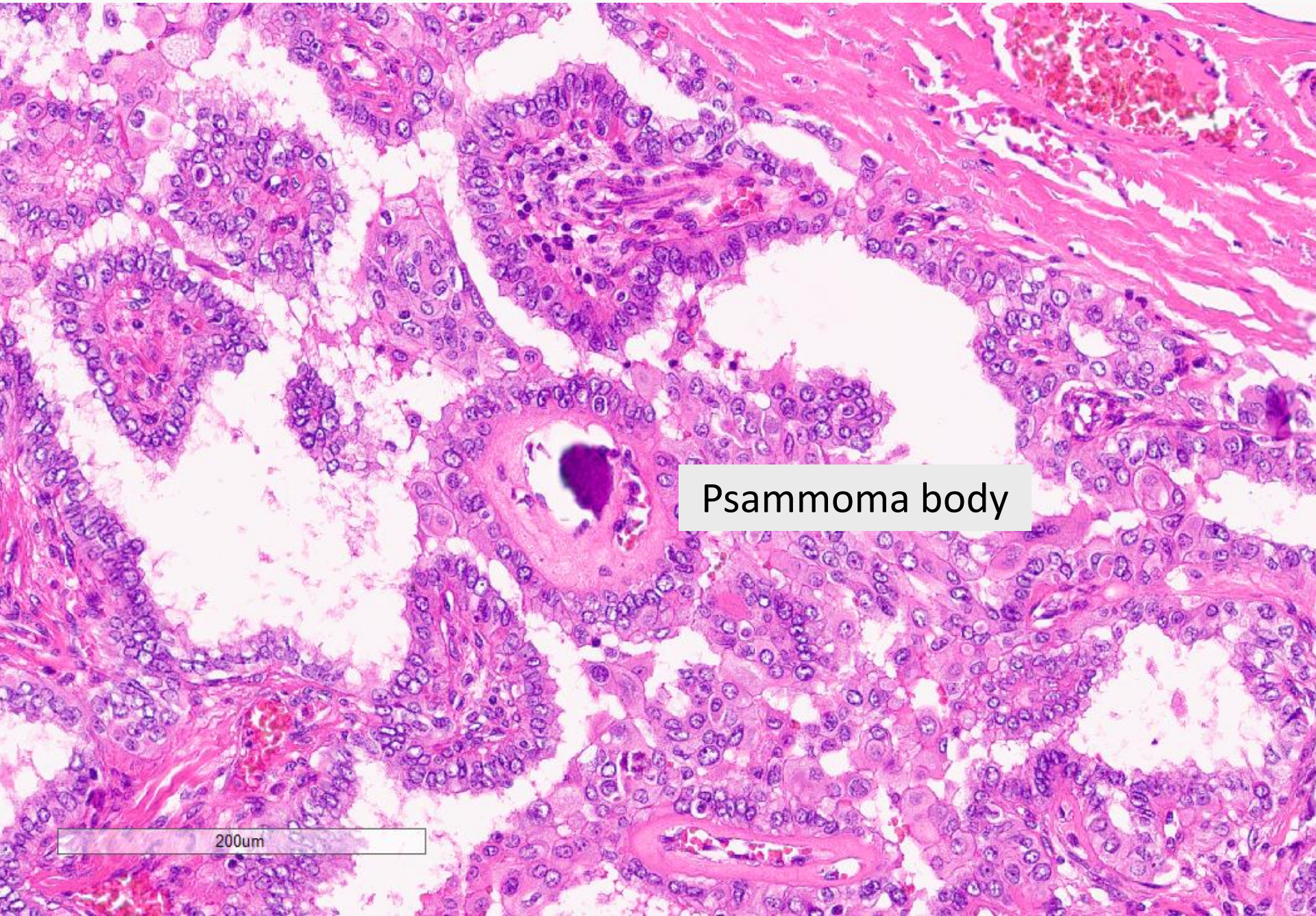
Classic pattern: complex, branching, randomly oriented papillae with fibrovascular cores

Papillary nuclear features

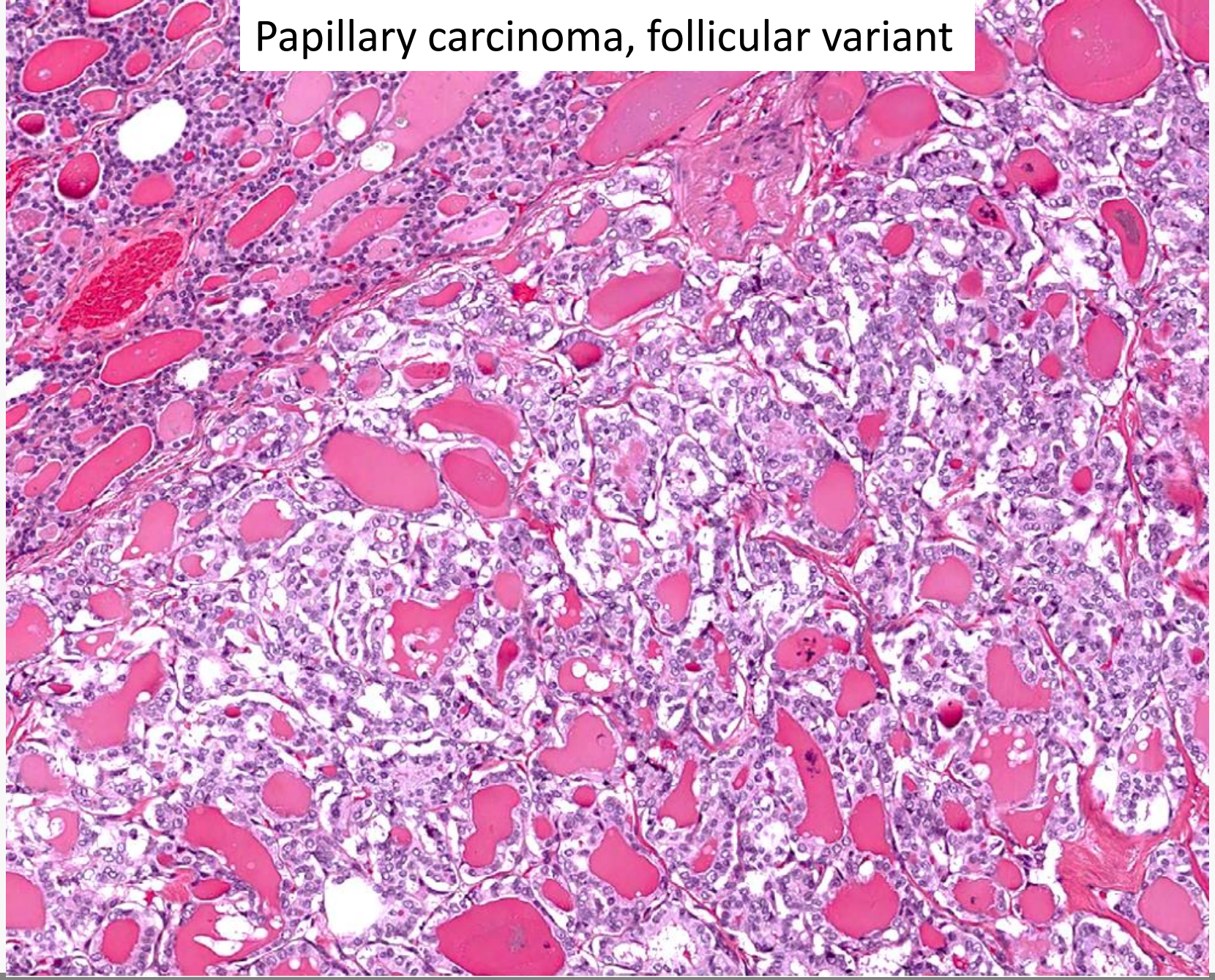



Nuclear groove
Nuclear groove

Papillary Carcinoma



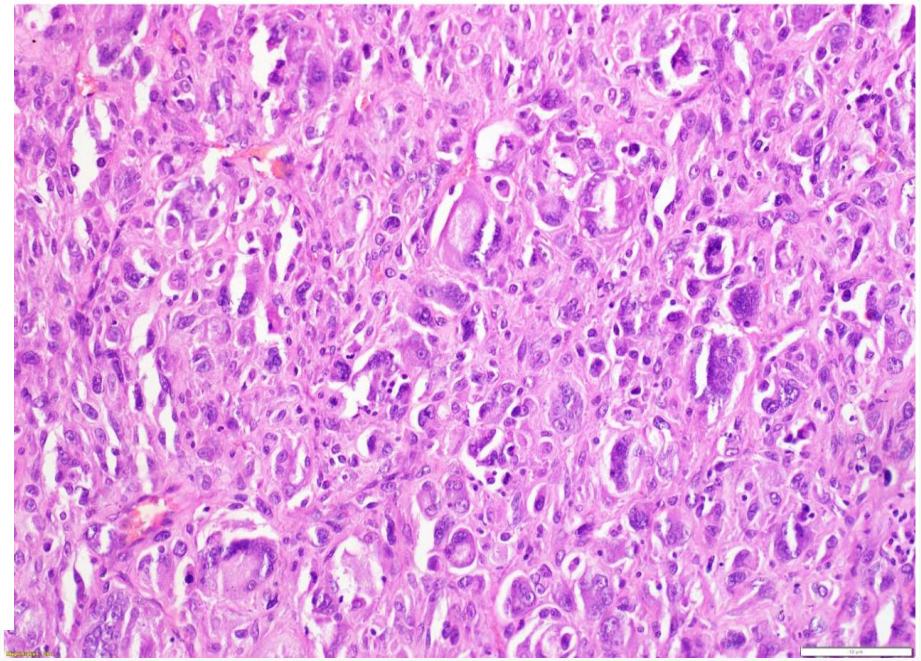
Papillary carcinoma, follicular variant



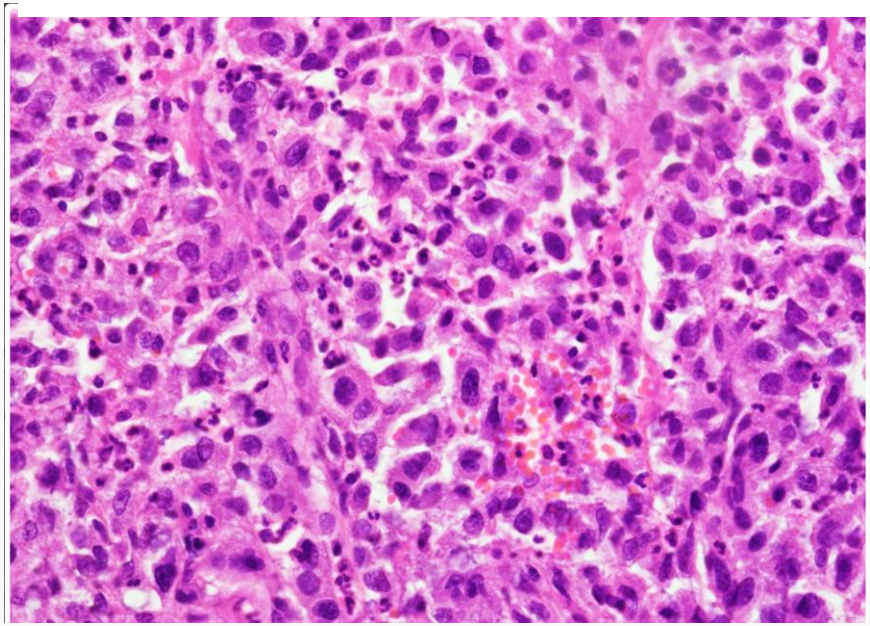
Undifferentiated carcinoma (anaplastic carcinoma)

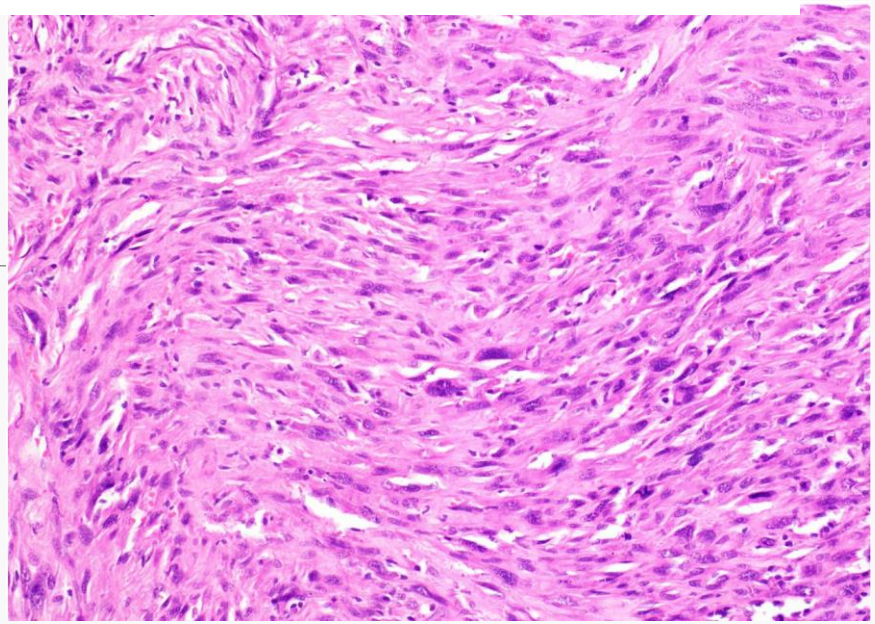
- Rare thyroid cancer
- Age >50 yrs, median 68 yrs
- Highly aggressive
- Rapidly enlarging fixed neck mass with compression symptoms (e.g., pain, dysphonia, dysphagia, dyspnea, cough)
- Poor prognosis,

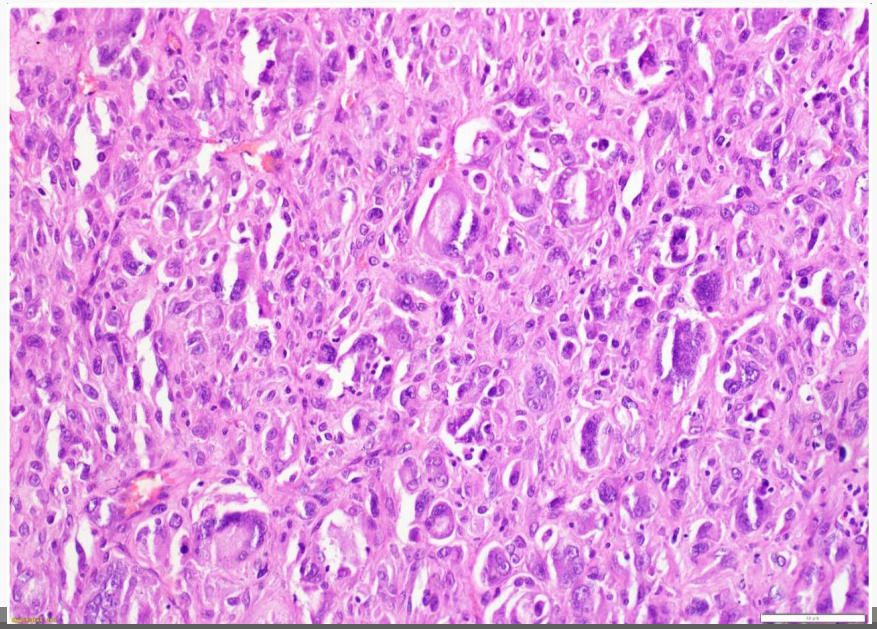
  Median survival: 5 - 6 months



Undifferentiated carcinoma (anaplastic carcinoma)



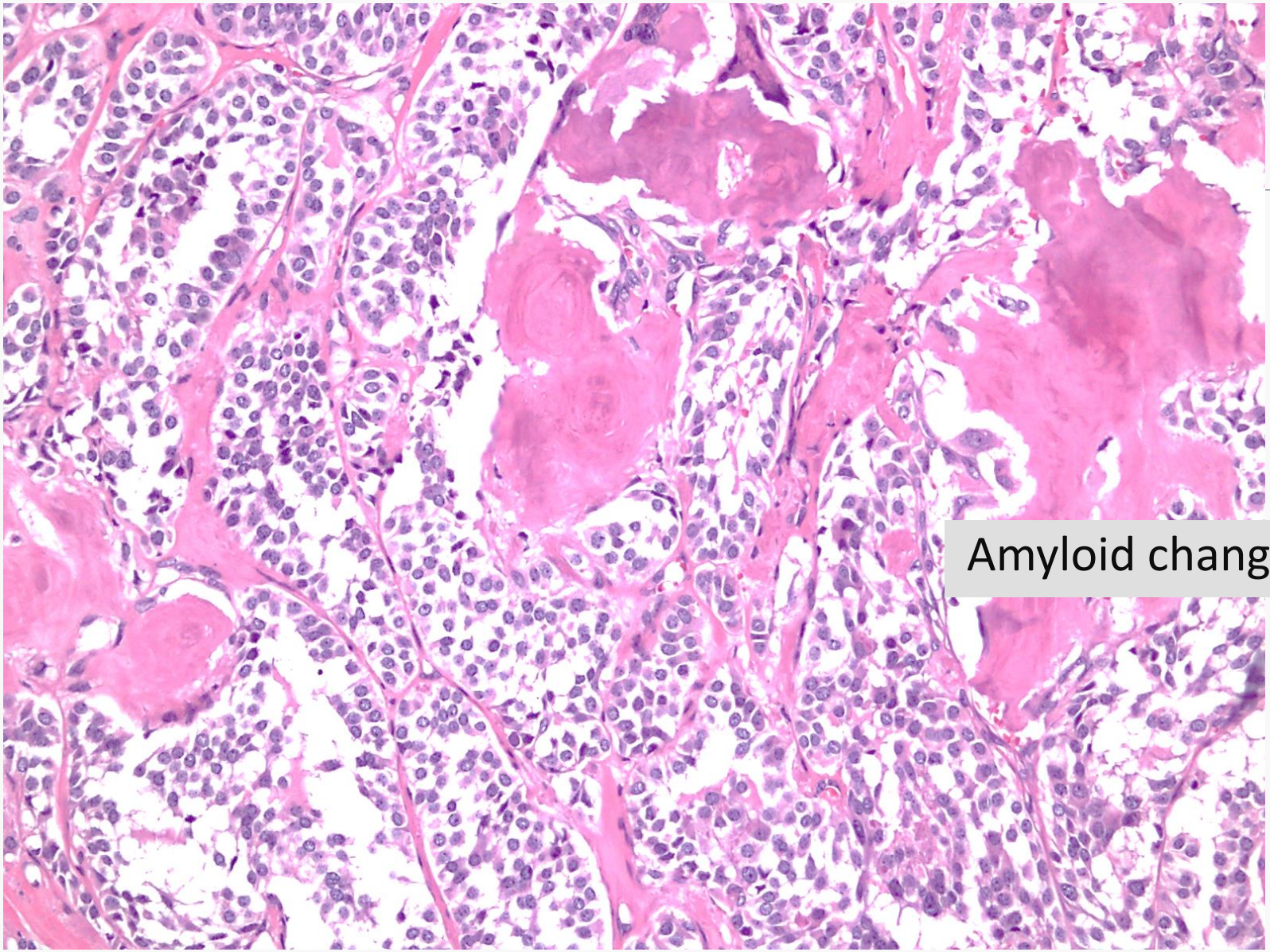




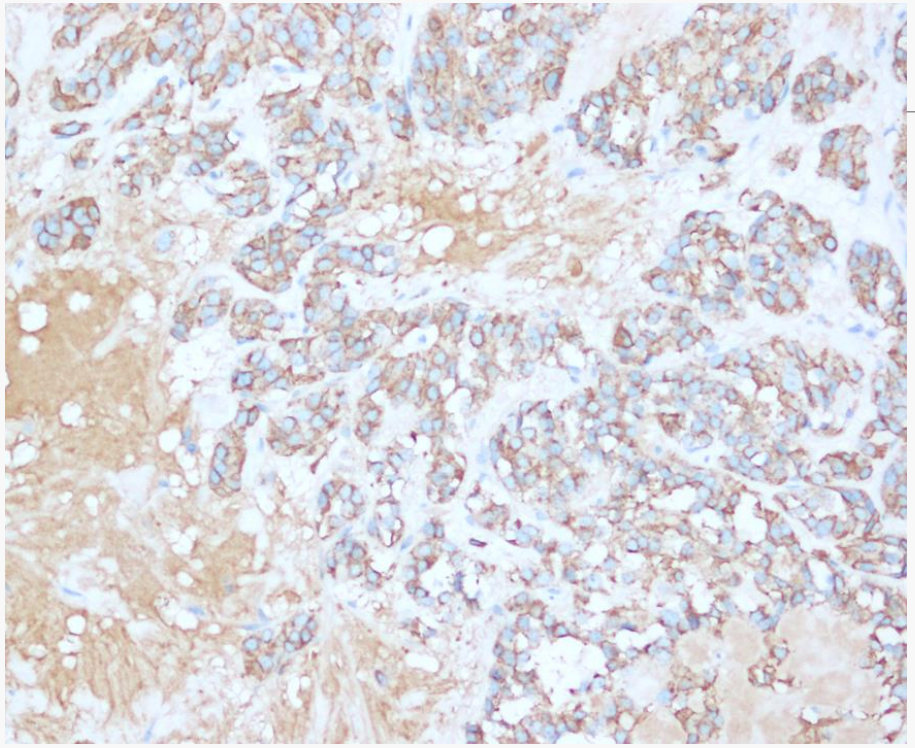
Histology:

- Marked cellular pleomorphism
- Bizarre, hyperchromatic nuclei
- Prominent mitotic activity
- Common pattern:
  - polygonal/epithelioid
  - spindle, giant cells

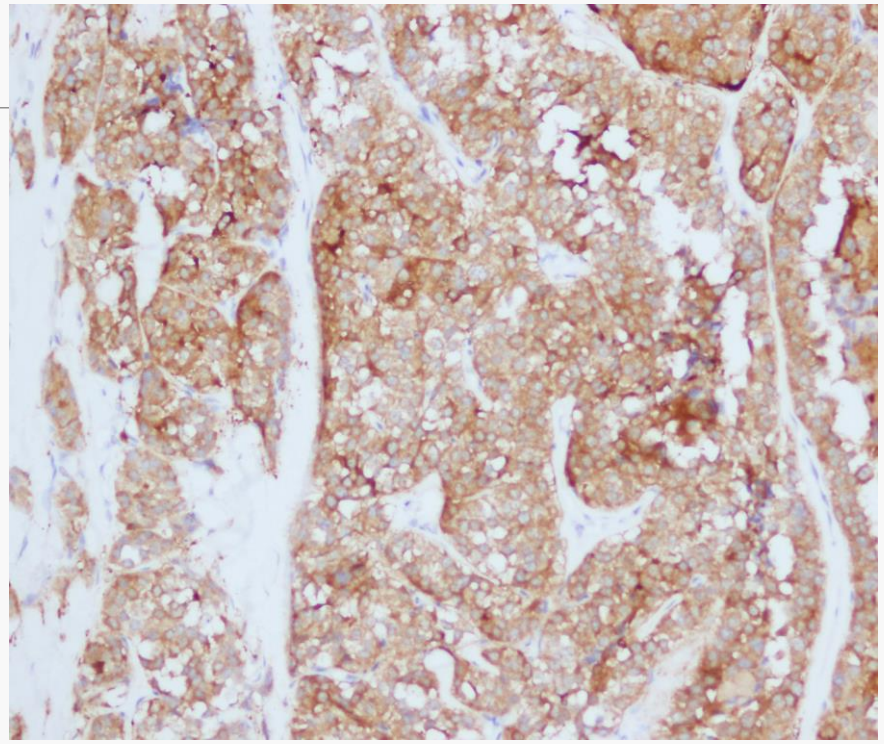
Medullary Carcinoma (C cell carcinoma)



Medullary Carcinoma (C cell carcinoma)



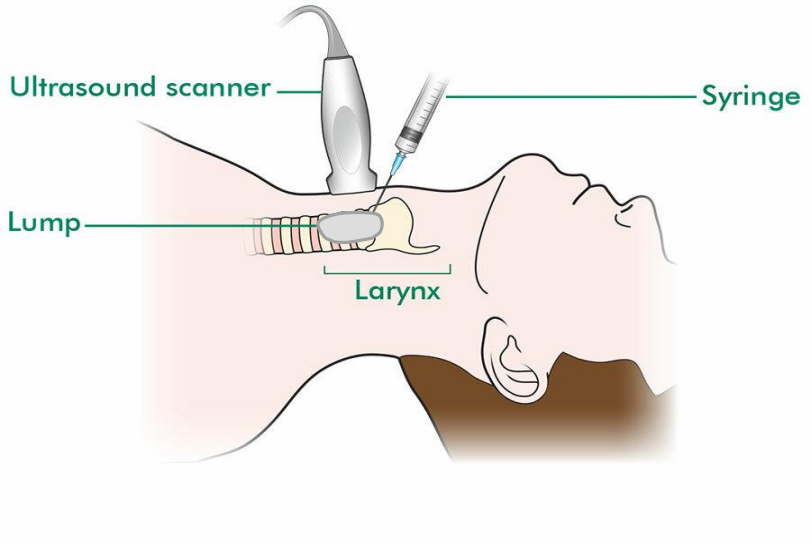
Chromogranin A



Neuron specific enolase (NSE)

Immunohistochemistry technique

Fine needle aspiration



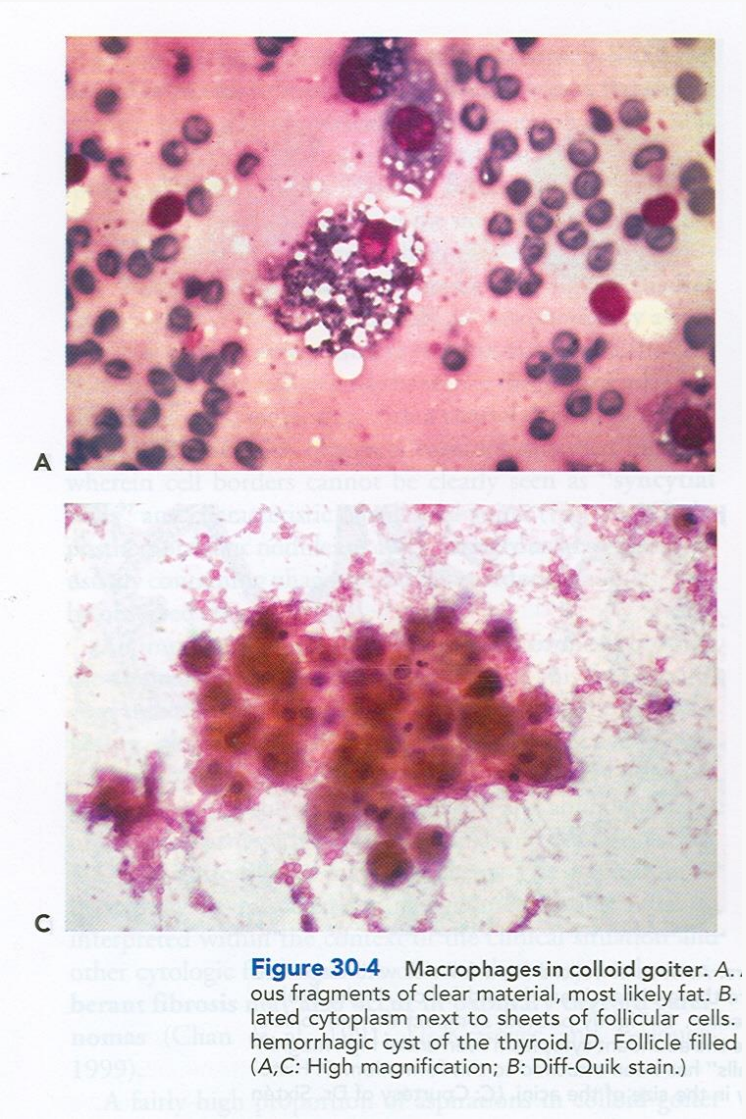
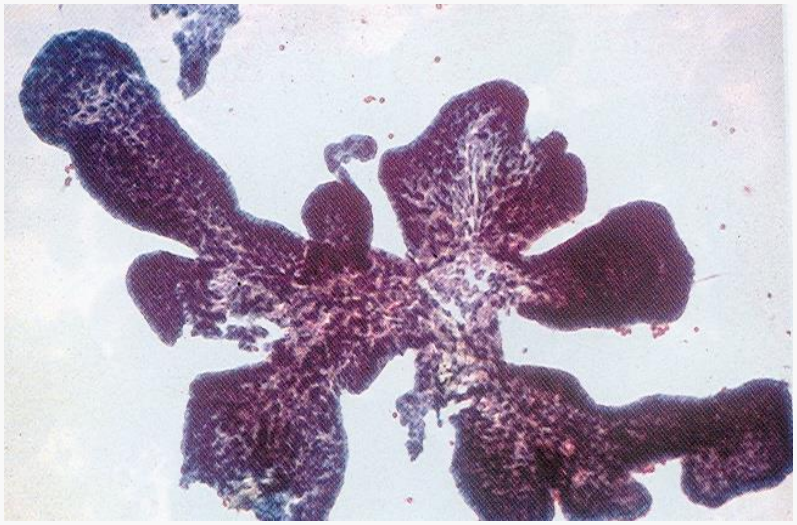
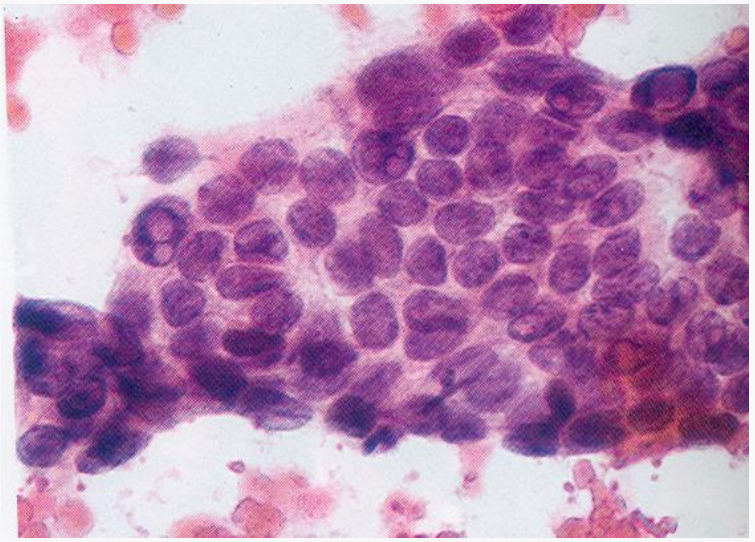
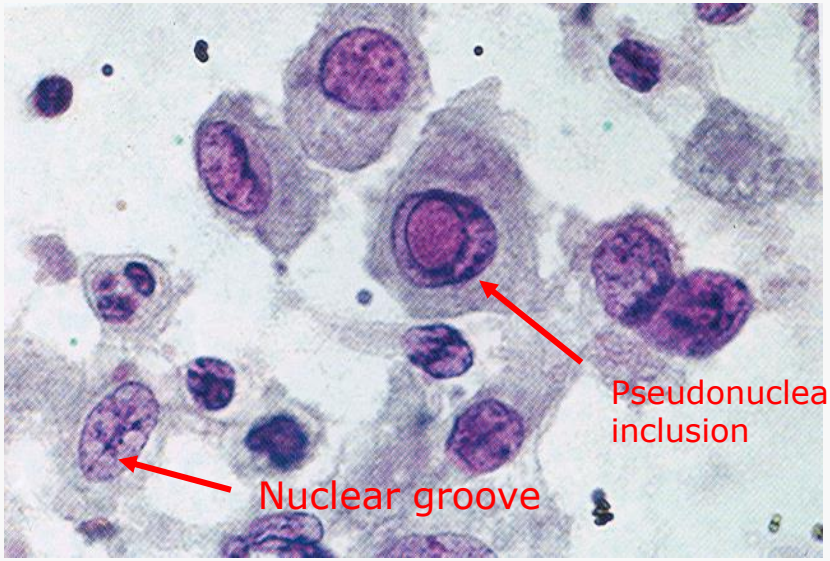


Figure 30-4 Macrophages in colloid goiter. *A.* ous fragments of clear material, most likely fat. *B.* lated cytoplasm next to sheets of follicular cells. hemorrhagic cyst of the thyroid. *D.* Follicle filled (*A,C:* High magnification; *B:* Diff-Quik stain.)

Fine needle aspiration

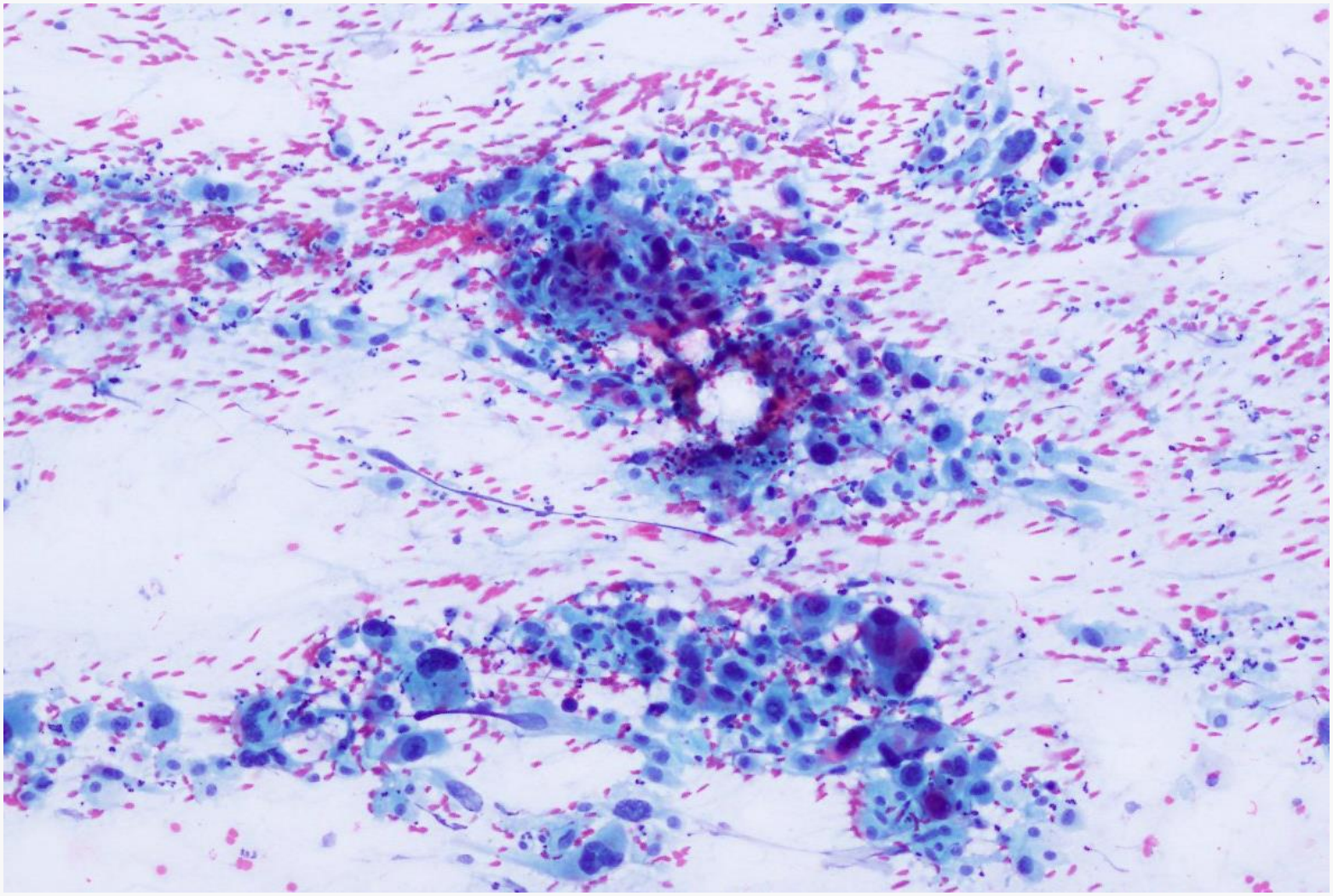






Papillary carcinoma

Fine needle aspiration



Undifferentiated carcinoma

Parathyroid gland

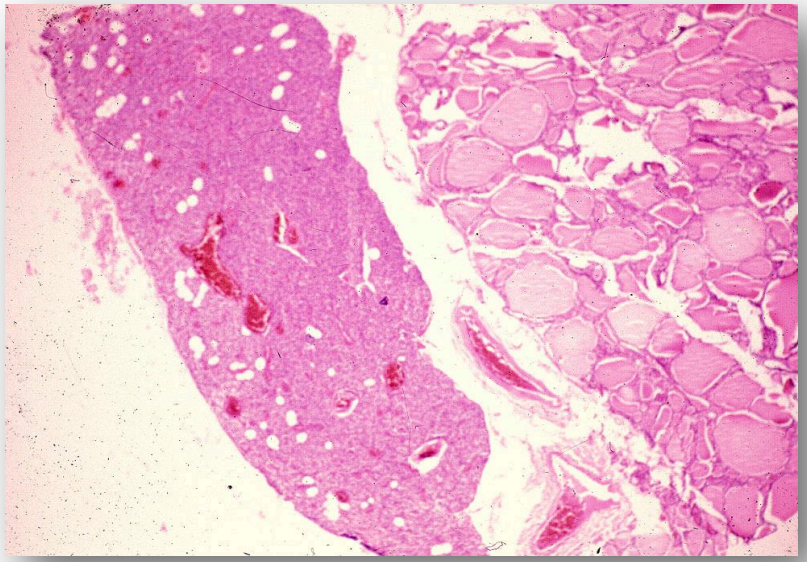
Parathyroid gland

พัฒนาจากกลุ่มเซลล์ที่บุ branchial pouch คู่ที่ 3, 4

ปกติมี 2-6 ต่อม แต่ส่วนใหญ่มี 4 ต่อม

Location: ด้านหลังของ thyroid gland

Size: 5x3x1 mm



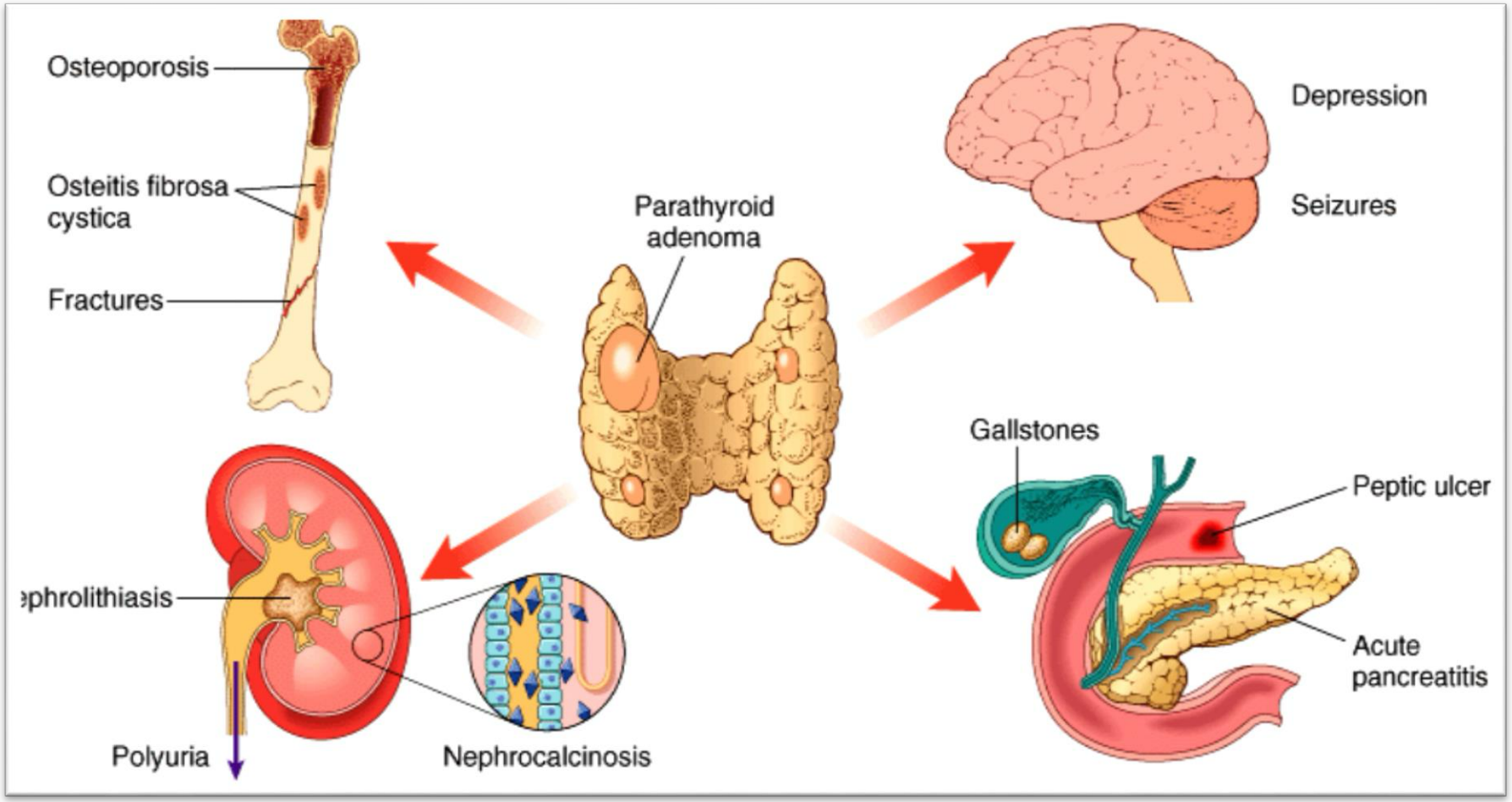
Normal parathyroid gland

Parathyroid gland

Function

- ดูดซึม Ca จากลำไส้เล็ก
- ละลาย Ca จากกระดูก
- ดึง Ca จากไต
- ขับ phosphate ออกจากไต

Cardinal features of hyperparathyroidism



อาการ

- มีนิ่ว, คลื่นไส้ อาเจียน เบื่ออาหาร ท้องอืด ซึมคล้ายโรคจิต
- Osteoperosis

Primary hyperparathyroidism

Causes:

- Parathyroid adenoma (most common)
- Primary hyperplasia,
- Parathyroid carcinoma (rare)

Lab:

- high PTH,
- high serum calcium,
- low phosphate

Secondary hyperparathyroidism

- Kidney ขับ phosphate ไม่ได้เนื่องจาก chronic renal failure
- Serum phosphate ที่สูง กระตุ้นให้ต่อมมี hyperplasia และหลั่ง PTH ออกมามาก

- Lab: High phosphate, very high PTH, low or normal serum calcium

Tertiary hyperparathyroidism

- Observe in patients with chronic secondary hyperparathyroidism and often **after renal transplantation**

- Parathyroid glands remain hyperfunctioning
  --> causing high PTH and normal calcium
  or high calcium

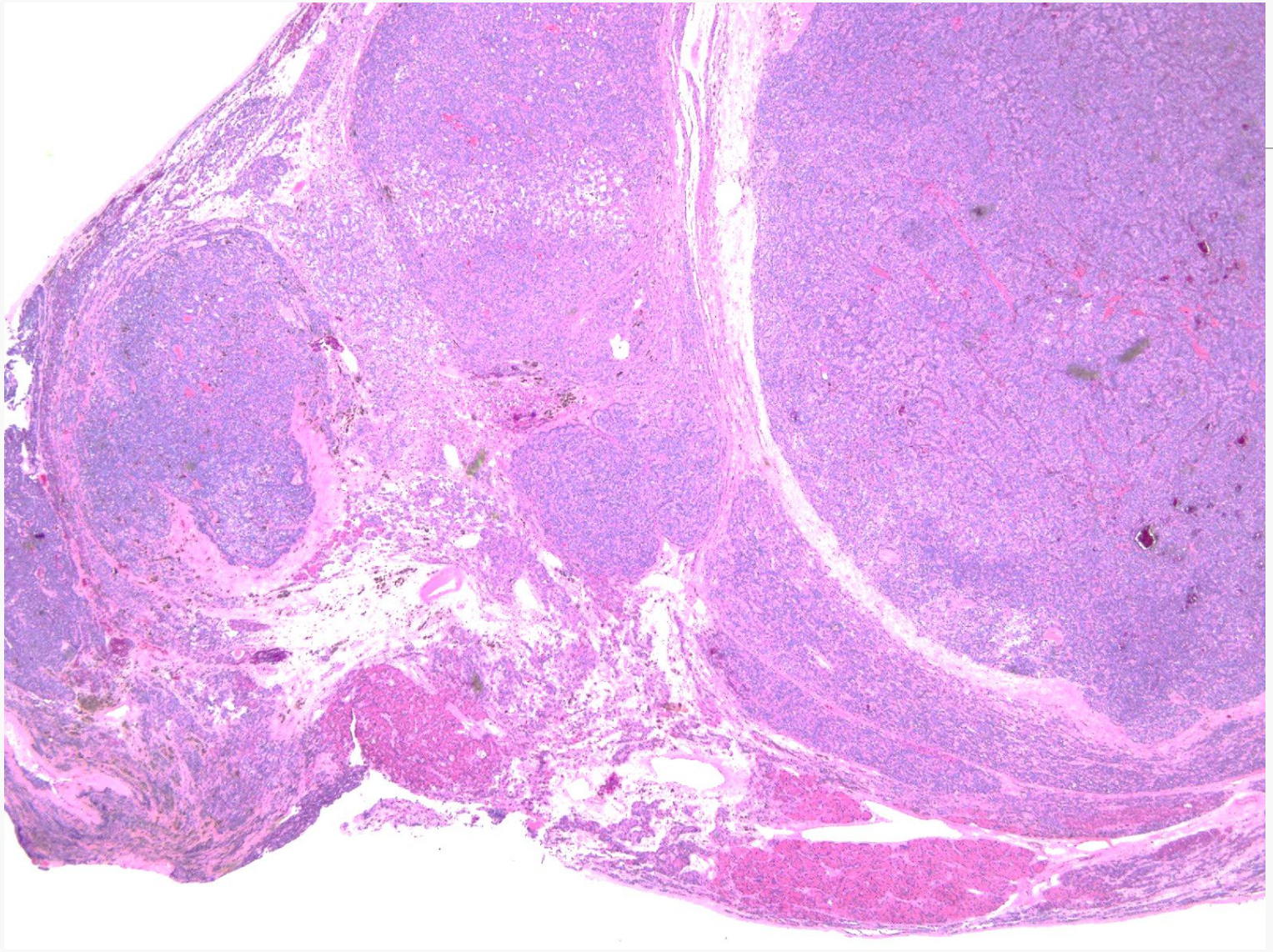
Parathyroid gland hyperplasia in patient with chronic renal failure



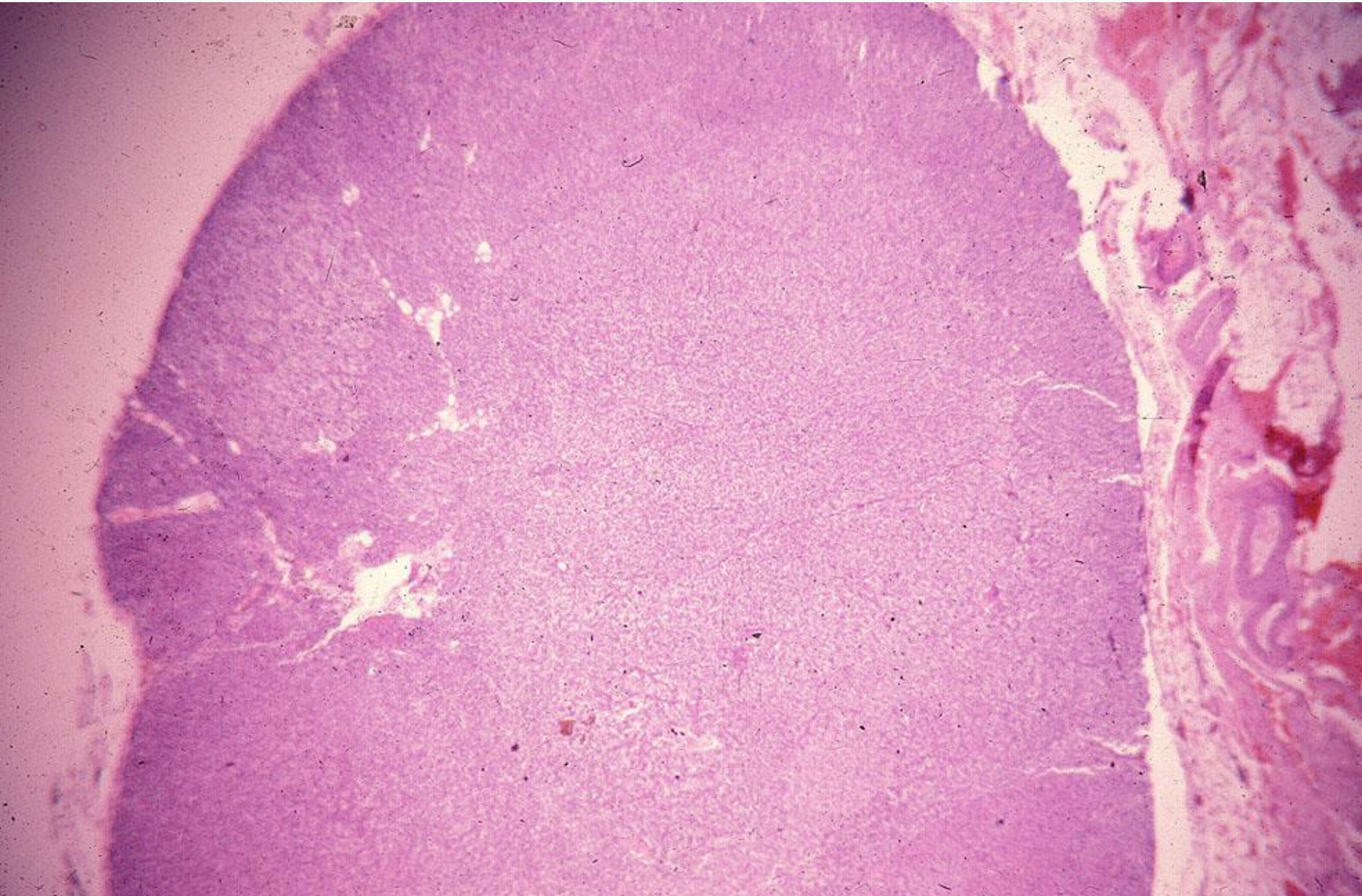
All four glands are enlarged with variation in size

https://www.pathologyoutlines.com/topic/parathyroidpthhyper.html

Parathyroid gland hyperplasia in patient with chronic renal failure



Parathyroid adenoma



Hypoparathyroidism

Decreased parathyroid hormone => hypocalcemia

Symptoms

: numbness, paresthesia, muscle cramping & twitching

Cause

- Surgery of thyroid or parathyroid gland
- Autoimmune
- Genetic


ANY
QUESTIONS?